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Non-pollen palynomorphs (NPPs) and their role in reconstructing the paleoclimate of the Seki-Eceler Höyük during the Holocene in southwestern Türkiye

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ABSTRACT

This study investigates non-pollen palynomorph (NPP) assemblages from the 4.7 m long JARK014-4 sediment core to reconstruct Holocene palaeoenvironmental changes at Seki-Eceler Höyük in southwestern Türkiye. During the early Holocene, the abundance of NPPs related to freshwater, including *Azolla*, *Spirogyra* and *Zygnema*, indicates the presence of extensive aquatic and wetland habitats. In addition, variations in *Glomus* which is an arbuscular mycorrhizal (AM) fungi, reflect changes in terrestrial and catchment inputs. Between 8200 and 7700 yr BP, the decline in aquatic NPPs and the increase in *Glomus* and *Scleroderma* are consistent with reduced freshwater availability and increased catchment disturbance associated with the 8.2 ka climatic event. However, the influence of hydrological controls cannot be ruled out. During the middle Holocene, *Sporormiella* type indicates the presence of herbivores and possible grazing activity. Furthermore, *Thecaphora* and *Tilletia* suggest an increase in the disturbance of herbaceous vegetation and possible agricultural activity. The late Holocene is characterized by a decline in freshwater-related NPPs and an increase in terrestrial fungal indicators, suggesting an increase in catchment disturbance due to interacting hydroclimatic variability and anthropogenic pressure. Numerous unidentified morphotypes are catalogued as EIES-015–EIES-165. Although numerous taxonomically unresolved morphotypes are documented as a morphological reference for future NPP studies within the EIES series, they are not used as primary paleoenvironmental indicators.

1. Introduction

The archaeological site of Seki-Eceler Höyük is located in the southwestern Türkiye (Fig. 1), and is characterized by a Mediterranean climate and vegetation. It has been largely overlooked in terms of both its natural and human history, resulting in few archaeological and geomorphological investigations (Aksoy and Köse, 2005; Momigliano et al., 2011). The rapid transition in topography from coastal to high, mountainous continental areas (Bakker et al., 2012) results in micro-environments with high biodiversity, which have attracted numerous civilizations throughout history to the region (Eastwood et al., 1999; Médail and Quezél, 1999). NPPs can improve the environmental accuracy of traditional pollen-based reconstructions by offering insights into aquatic conditions, catchment disturbance, the presence of herbivores, and other ecological processes that cannot be easily identified from pollen data alone. NPPs have applications in archaeology, geology, and palaeoclimatology (Limaye et al., 2007). While pollen primarily reflects vegetation composition and change, NPP assemblages can provide additional evidence of hydrological variability, erosion, aquatic ecosystem dynamics and grazing-related disturbances (Menozzi et al., 2010). However, responses of NPPs to rapid environmental or geological disturbances are generally indirect, via associated changes in hydrology, erosion or sediment input. Therefore, they should be interpreted alongside independent sedimentological or geochemical

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evidence. NPP analysis has become a well-established addition to palynological and multiproxy studies (Van Geel and Van der Hammen, 1978; Miola, 2012; Shumilovskikh et al., 2021). In archaeological and palaeoecological studies, coprophilous fungal spores are widely used to assess the presence of herbivores and the pressure they exert through grazing. However, these indicators cannot distinguish between domestic and wild herbivores, so they are most reliable when considered alongside pollen, archaeological and other evidence. More broadly, NPP assemblages have been successfully used to reconstruct changes in the hydrology, ecology and human–environment interactions in a variety of depositional settings (Mighall et al., 2006; Limaye et al., 2007; Tunno and Mensing, 2017). NPP assemblages comprise a variety of microscopic organic remains, such as fungi, algae, cyanobacteria, testate amoebae, and plant and zoological remains, which are recovered during palynological processing (Shumilovskikh et al., 2021). Their composition is influenced by ecological conditions, depositional factors, taphonomic factors and laboratory processing factors, all of which must be considered when assigning palaeoenvironmental significance (Limaye et al., 2007; Medeanic and Silva, 2010; van Asperen et al., 2021; Shumilovskikh et al., 2021). In the present study, fungi and freshwater algae are particularly useful because they provide complementary information on catchment disturbance, herbivore activity, and aquatic conditions. For instance, *Glomus* reflects catchment erosion and terrestrial input, whereas freshwater taxa such as *Zygnema* and *Spirogyra* provide insights into wetland habitats (Medeanic and Silva, 2010; Gelorini et al., 2011). Nevertheless, such ecological associations are interpreted at the assemblage level rather than treating individual taxa as definitive indicators of a single environmental driver. Despite the increasing use of NPP analysis, long-term NPP records in Türkiye remain comparatively limited. NPP-based palaeoecological investigations have been done at various locations, including Lake Manyas (Kutluk, 2019), Lake Hazar (Biltekin et al., 2021; Öntürk and Kutluk, 2022), the Bay of Elaia (Seeliger et al., 2017), Lake Sülük (Biltekin et al., 2025) and Lake Marmara (Kolbüken et al., 2025). Alongside these site-specific studies, the NPP-ID database is a valuable resource for identifying and comparing the taxonomy of NPPs (Shumilovskikh et al., 2022). Other microfossil studies, including investigations of dinoflagellate cyst distributions in surface sediments, also contribute to the regional palaeoenvironmental framework (Gurdebeke et al., 2018).

Nevertheless, continuous Holocene NPP records from archaeological settings in southwestern Türkiye remain scarce. The main objective of this study is therefore to analyse NPP assemblages from core JARK014-4 in order to reconstruct Holocene changes in terms of freshwater availability, aquatic habitats, catchment disturbance and terrestrial inputs at Seki-Eceler. A second objective is to evaluate possible human–environment interactions, including the presence of herbivores, grazing-related disturbance and potential agricultural activity, where these interpretations are supported by independent pollen and archaeological evidence. Finally, the study documents a diverse suite of EIES-coded morphotypes, distinguishing between confidently identified taxa and those with only tentative taxonomic affinities, as well as currently unidentified forms. The latter are presented primarily as a morphological reference and are not used as standalone palaeoenvironmental indicators.

1.1. Study area and geology

The Seki-Eceler Höyük (36°48′52.1″N, 29°38′13.1″E; see Fig. 1) s an archaeological settlement located in the Seki Basin of southwestern Türkiye, at approximately 1100 m a.s.l. The site lies within a high-altitude intramontane setting surrounded by the Western Taurus Mountains. Geologically, the Seki Basin is situated within the regionally extensive tectonostratigraphic complex of southwestern Türkiye (Fig. 2). The archaeological site is located on the ophiolitic mélange (Danelian et al., 2006), while the surrounding terrain comprises units belonging to the Lycian Nappes (Collins and Robertson, 1998; Danelian et al., 2006; Alçiçek et al., 2015; Keserci et al., 2023). These geological units form the surrounding catchment and therefore represent potential sources of detrital material transported into the basin through erosion and fluvial processes. The region has a Mediterranean climate with mild, wet winters and long, hot summers (Özhatay et al., 2013; Lionello et al., 2022; Tüfekcioğlu and Tavşanoğlu, 2024). At this relatively high elevation, however, the environmental conditions are also strongly influenced by topography and orographic effects. Hydroclimatic

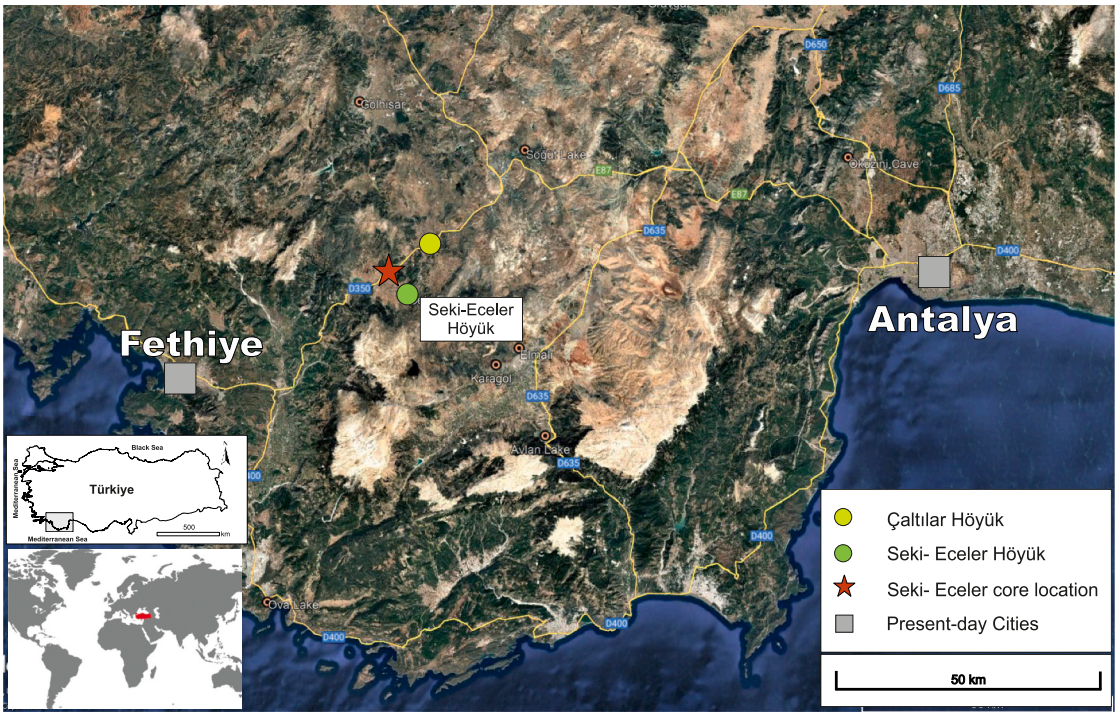


Fig. 1. Location map of the Seki-Eceler Basin.

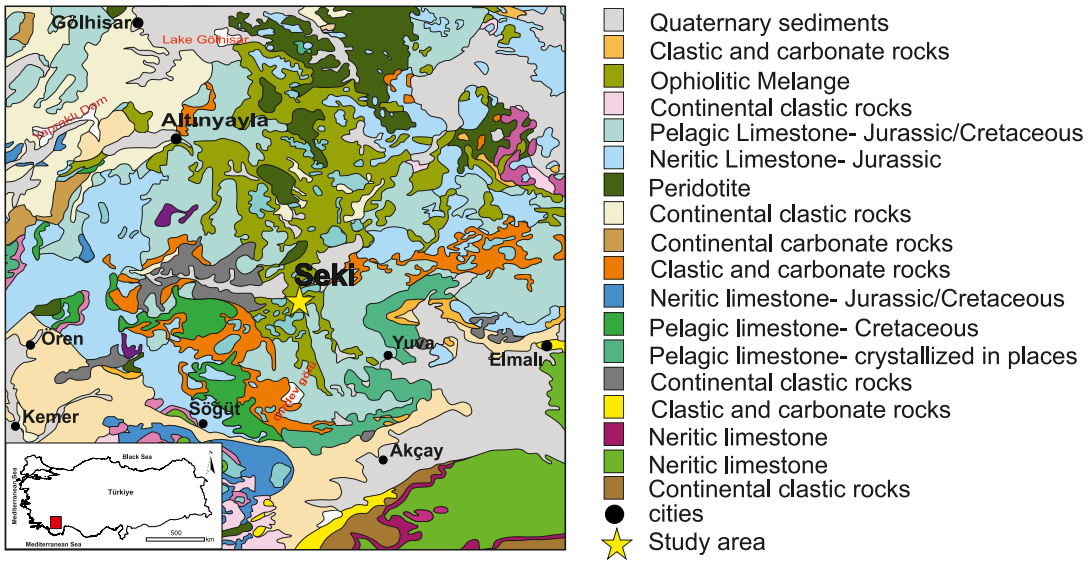


Fig. 2. Geological map of the study area (modified from Hashemi et al., 2026a,b, and the 1:500,000 Scale Geological Inventory Map Series of Türkiye prepared by MTA). https://www.mta.gov.tr/en/sayfalar/maps/images/1-500/DENIZLI.jpg.

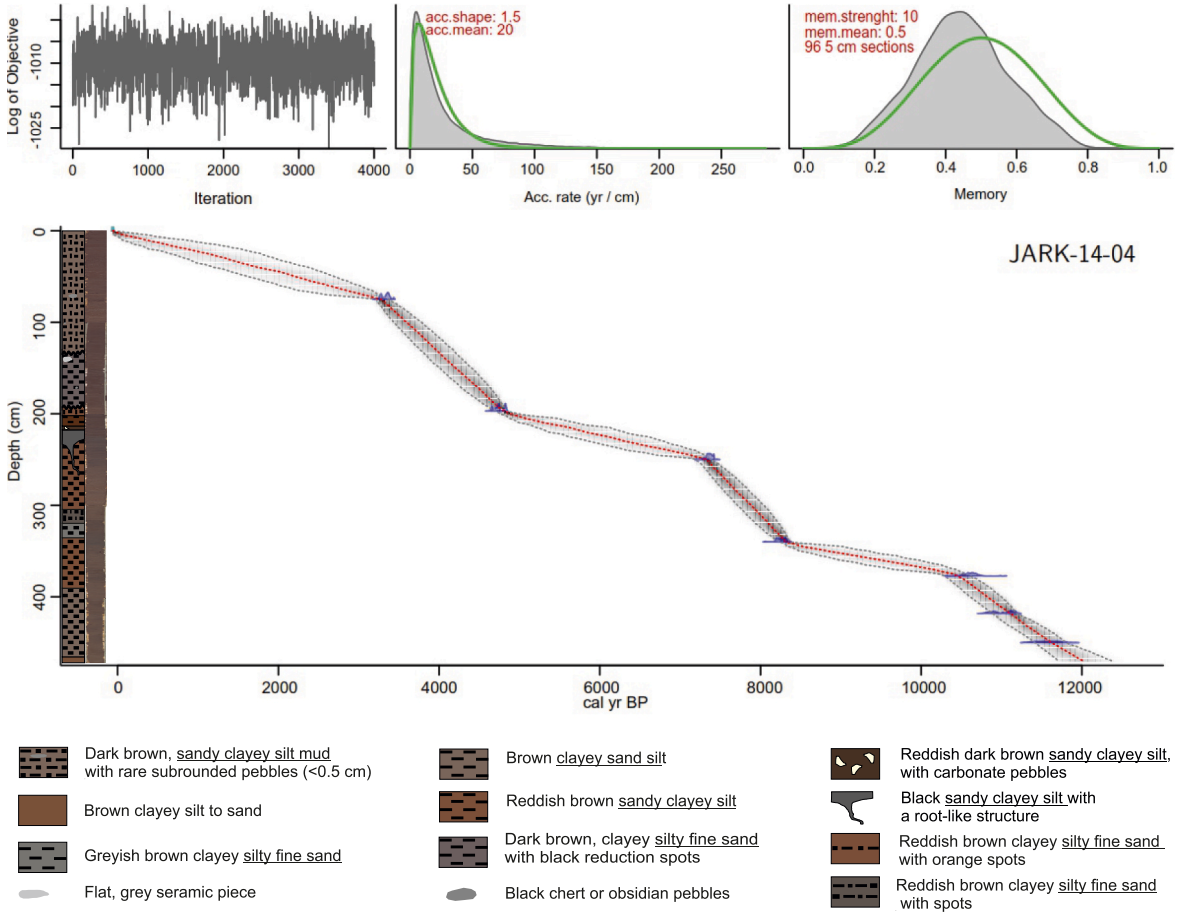


Fig. 3. Age-depth modeling and lithological log with optical image of the core JARK14−04.

variability in the broader Mediterranean region is associated with atmospheric circulation patterns including the North Atlantic Oscillation (NAO) and Mediterranean Oscillation (MO) (Cenk and Turgay, 2020; Criado-Aldeanueva and Soto-Navarro, 2020; Rowe et al., 2020). Its long settlement history, dating back to the late Neolithic/early Chalcolithic period (Aksoy and Köse, 2005; Momigliano et al., 2011), makes this location one of the most valuable sites for archaeological, palynological and paleoecological studies.

2. Materials and methods

2.1. Coring, sampling and sedimentological studies

The JARK14–04 sediment core was obtained from the floodplain of the Seki Stream, adjacent to the Seki–Eceler Höyük site in southwestern Türkiye, using a vehicle-mounted hydraulic drilling rig. A continuous sediment sequence, measuring 4.7 m in length, was recovered. The coring location was selected in consultation with archaeologists to ensure that it was situated outside the protected archaeological zones while providing a sedimentary record representative of environmental conditions in the vicinity of the Höyük. Following recovery, the core was transported to the EMCOL Applied Research Centre at Istanbul Technical University (ITU), where it was opened, visually described, logged, and sampled. The lithological characteristics of the entire recovered sequence were documented using vertical stratigraphic logging, with particular attention to sediment texture, lithological transitions, and the boundaries between the main sedimentary units. Visual documentation of the core, together with the corresponding lithological log, is provided in Fig. 3. A total of 42 samples were collected systematically at 10 cm intervals along the continuous sediment sequence at the ITU–EMCOL Applied Research Centre.

2.2. Age-depth modeling chronology

Ten bulk sediment samples from core JARK14–04 were sent to the Poznań Radiocarbon Laboratory for accelerator mass spectrometry (AMS) radiocarbon dating (Table 1, Fig. 3). The chronology used in the present study represents a revised version of the age-depth model previously presented by Hashemi et al. (2026a,b). The model was reassessed to provide a more transparent treatment of chronological uncertainty and of radiocarbon determinations that were inconsistent with the general stratigraphic sequence.

All radiocarbon determinations were calibrated using IntCal20 (Reille, 1992) and Bayesian age-depth modelling was performed with rbacon (Blaauw and Christen, 2011) in R (Core Team, 2024). The model was also constrained using a core-top calendar-age tie point of −60 cal yr BP. Sedimentation priors were kept at their default settings, except for memory, for which a relatively high prior mean was specified to reflect the continuous accumulation of the recovered sediment sequence. For the three potentially anomalous radiocarbon determinations, the Student's t-distribution was specified with 30 degrees of freedom (df = 30). This value was selected to strike a balance between a near-normal likelihood and a more heavily tailed distribution. This allows for greater tolerance of stratigraphically inconsistent determinations, while preventing individual anomalous ages from having an excessive influence on the modelled chronology.

We acknowledge that fixing the degrees of freedom parameter affects the width and shape of the likelihood distribution, and consequently may impact posterior chronological uncertainty. In particular, a relatively high df value produces a distribution closer to a normal distribution and may yield narrower uncertainty than a more heavily tailed model. Therefore, the resulting credible intervals should be interpreted as being conditional on the specified model structure and priors, rather than as a complete representation of all possible sources of chronological uncertainty. Posterior ages are reported as medians with 89% highest posterior density intervals. These intervals represent the uncertainty propagated by the Bayesian age-depth model under the specified calibration curve, likelihoods, accumulation and memory priors, and chronological constraints. Ages below the deepest dated level were extrapolated using Bacon's prior accumulation structure (Fig. 3).

Table 1
AMS radiocarbon and calibrated ages of core JARK14–04 (Hashemi et al., 2026a,b). All measurements are on bulk sediment total organic carbon (TOC). Ages are reported as radiocarbon yr BP (relative to 1950 CE) ±1σ. The last column lists the per-date Student-t likelihood hyperparameters (t.a, t.b) used in rbacon to reflect differing reliability: (33, 34) approximates a near-normal likelihood, whereas (3, 4) is heavy-tailed to down-weight potential outliers. No per-sample calibrated ages are given here; calibration is performed only within the Bayesian age-depth model (see Methods). A calendar-age tie point at the core top (0 cm, −60 cal yr BP, σ = 10 yr) was applied separately and is not included in this table. TOC stands for Total Organic Carbon.

Lab code	Dated material	Core depth (cm)	^{14}C years BP	(t.a, t.b)
Poz-170,065	TOC	75	3125 ± 35	(33, 34)
Poz-170,066	TOC	102	4210 ± 30	(3, 4)
Poz-169,999	TOC	197	4210 ± 30	(33, 34)
Poz-95796	TOC	210	4005 ± 35	(3, 4)
Poz-95794	TOC	250	6460 ± 40	(33, 34)
Poz-170,067	TOC	309	2300 ± 30	(3, 4)
Poz-95768	TOC	340	7430 ± 50	(33, 34)
Poz-170,228	TOC	377	9390 ± 60	(33, 34)
Poz-170,095	TOC	418	9660 ± 60	(33, 34)
Poz-95767	TOC	450	10,050 ± 70	(33, 34)

2.3. Palynological analysis

To perform a high-resolution palynological analysis, sediment samples were collected at 10 cm intervals and processed in accordance with standard protocols (Cour, 1974). The same chemically treated residues were used for both pollen and NPP analyses. This included the use of cold HCl and HF to remove carbonates and silicates, respectively. Zinc chloride ($ZnCl_2$) was then employed to separate palynomorphs with a density greater than 2 g/cm^3, after which the samples were sieved through 200 μm and 10 μm mesh. The upper 200 μm sieve was used to remove coarse sedimentary and organic material, whereas the 10 μm sieve was used to remove the finest particles. The pollen residues were mounted in glycerine and analysed via transmission light microscopy at 63× and 100× magnification (with oil immersion). Non-pollen palynomorphs (NPPs), which vary in size from approximately 10–250 μm, can be observed during routine pollen counting procedures (Shumilovskikh and van Geel, 2020; Shumilovskikh et al., 2015, 2016, 2019, 2021, 2022). Percentages were calculated based on the total NPPs count. A totally, we counted 173 NPPs. Of these, 23 were identified and 150 were unidentified and coded as EIES in core JARK14–04. Identification of the non–pollen palynomorphs (NPPs) was performed via descriptions and illustrations from various sources, including Wolf (1966, 1967), Van Geel and Van der Hammen (1978), van Geel et al. (1989), van Geel (2001), Ellis and Ellis (1997), Ellis-Adam and van Geel (1978), van Geel and Aptroot (2006), Cugny et al. (2010), Cugny (2011) and Miola (2012), as well as the Non–Pollen Palynomorph Image Database (https://non-pollen-palynomorphs.uni-goettingen.de/). This extensive referencing ensures the accurate classification and comparison of NPPs across different studies, providing a comprehensive understanding of their distribution and ecological significance. The findings are presented in the form of comprehensive non-pollen diagrams, which were generated using the TiliaIT programme (version 2.0.41; Grimm, 1991–2015). To identify non-pollen zones, TiliaIT includes a CONISS (Constrained Incremental Sum of Squares) analysis that considers changes in non-pollen species (Grimm, 1987).

The NPP assemblages were classified into three levels of taxonomic confidence: (1) identified taxa with well-established morphological and ecological affinities; (2) morphotypes showing probable or tentative affinities to previously described taxa or morphological groups; and (3) unidentified EIES-coded morphotypes for which no reliable taxonomic assignment could be made. Only the first category was used as primary indicators in the paleoenvironmental reconstruction, and only those with well-established ecological significance. Morphotypes that were tentatively assigned were only used for morphological comparison and assemblage characterization, whereas unidentified EIES morphotypes were excluded from environmental interpretations.

3. Results

3.1. Chronology and lithology

Based on the age-depth model, the 4.7 m long JARK014-4 core provides a high-resolution stratigraphic record of the Seki-Eceler Höyük region. The lowermost section of the core, dated to approximately 12,000 yr BP, corresponds to the Younger Dryas period. It is characterized by a layer of brown, clayey, sandy silt. The transition from the Younger Dryas to the early Holocene is marked by the deposition of this layer at a depth of 450 cm (Fig. 3). The core succession then continues with a reddish-brown, sandy, clayey silt layer containing root-like structures. This layer accumulated during the transition from the early to middle Holocene. This layer is overlain by a reddish-dark brown, sandy, clayey silt layer that is rich in carbonate pebbles. A thin, reddish-brown, clayey, silty, fine sand layer containing distinctive 1–8 mm orange spots overlies this section. The top of this layer, at 197 cm, is dated to approximately 4800 cal yr BP. Above this, from 197 cm to 130 cm, the core consists of dark brown, clayey, silty fine sand containing black reduction spots, black chert or obsidian pebbles, and distinct layers of flat grey ceramics (Fig. 3). The upper boundary of this layer, at 130 cm, is dated to 3950 cal yr BP. The uppermost section of the core is a thick layer, approximately 130 cm deep, of dark brown, sandy, clayey silt containing rare subrounded pebbles. The topmost subsample, taken at a depth of 75 cm, is dated to 3300 cal yr BP. The mean accumulation rate for the core is estimated to be approximately 0.5 mm per year, or 20 years per cm.

3.2. Ecological classification and description of the non-pollen palynomorph assemblages

The JARK14–04 core contains a diverse assemblage of terrestrial, aquatic, fungal and zoological non-pollen palynomorphs. Descriptions of unidentified new non-pollen palynomorph records are given in Supplementary File (S1). Instead of describing each taxon individually, the identified non-pollen palynomorphs (NPPs) were grouped according to their principal ecological affinities and potential relevance to hydrology, catchment disturbance, trophic conditions and human activities (Table 2). In addition to the NPP taxa that could be confidently identified, numerous morphotypes without secure taxonomic assignments were documented in core JARK014–4 and catalogued using the EIES numbering system. These morphotypes were separated according to their level of taxonomic certainty. Some specimens show morphological similarities to previously described fungal or zoological groups and are reported as having probable or tentative affinities; others remain unidentified. Terms such as *Nigrospora*-like, *Sordaria*-type, dictyospore-like, phragmospore-like, amerospore-like, rotifer-like and chironomid/insect-like are used only to indicate morphological resemblance and should not be regarded as definitive taxonomic identifications. The EIES morphotypes are presented primarily as a morphological reference and to document NPP diversity in the Seki-Eceler sequence. They were not used as primary paleoenvironmental indicators unless a sufficiently secure taxonomic identification and well-established ecological relationship were available. Thefore, the paleoenvironmental interpretations presented below are principally based on taxa that have been identified with confidence, including *Glomus*, *Scleroderma*, *Sporormiella* type, *Azolla*, *Spirogyra*, *Zygnema*, *Arcella* and *Pediastrum*, as well as other NPPs that have been shown to be ecologically significant.

The terrestrial and catchment-related group is dominated by *Glomus* and *Scleroderma*. *Glomus* is associated with plant roots and can accumulate in sediments through soil erosion and catchment runoff (Gelorini et al., 2011; van Asperen et al., 2021). *Scleroderma* can also accumulate in this way, reflecting the erosion and transport of soil-derived fungal material (Dal Corso et al., 2019). The presence of fungal hyphae and Microthyriaceae provides additional information on terrestrial organic matter and humid conditions, although their abundance may also be influenced by preservation and transport processes. Indicators of aquatic and wetland environments include *Azolla*, *Spirogyra*, *Zygnema*, *Pediastrum*, *Ceratophyllum* leaf spinule, *Arcella*, *Plumatella* statoblasts and rotifer remains. Changes in these species provide information on the extent and characteristics of freshwater habitats. *Spirogyra* and *Zygnema* are associated with freshwater environments, whereas the presence of *Azolla* indicates shallow surface-water habitats. *Pediastrum*, rotifer eggs and filamentous cyanobacteria may also reflect changes in aquatic productivity and trophic conditions. High abundances of cyanobacteria and rotifer remains have been linked to eutrophic conditions (Revelles et al., 2016; McCarthy et al., 2021). A third ecological group comprises NPPs that are potentially associated with grazing, pastoralism and agricultural disturbance. *Sporormiella*-type spores are coprophilous and are commonly associated with the presence of herbivores and grazing activity (Davis and Shafer, 2006; Mighall et al., 2006; Loughlin et al., 2018). *Thecaphora* and *Tilletia* are associated with herbaceous plants and grasses, and may be found in areas affected by agriculture (Carrión and Navarro, 2002; Dal Corso et al., 2019). These taxa are considered supporting indicators rather than definitive evidence of specific land use practices.

The Pseudoschizaea (*Chomotriletes/Moyeria*-type) group is associated with wet soils, marsh environments and periods of hydrological instability or seasonal drying (Carrión and Navarro, 2002; Montoya et al., 2012; Revelles et al., 2016). Other remains, including monolete and trilete spores, Lepidoptera wing scales, HdV-989, *Peronospora*-type oospores and several zoological fragments, were also found in the

Table 2
Ecological classification and palaeoenvironmental significance of the main NPP groups identified in core JARK14–04.

Ecological group	Main NPP types	Principal ecological association	Potential palaeoenvironmental significance
Catchment/soil disturbance	*Glomus*, *Scleroderma*, fungal hyphae	Soils, plant roots, catchment-derived organic material	Catchment erosion, runoff and terrestrial disturbance
Humid terrestrial environments	Microthyriaceae	Epiphyllous fungi, humid vegetation	Locally humid conditions and vegetation-associated fungal activity
Freshwater/wetland	*Azolla*, *Spirogyra*, *Zygnema*, *Ceratophyllum*, *Arcella*	Shallow freshwater, ponds, wetlands and wet soils	Changes in freshwater availability and wetland extent
Aquatic productivity/ trophic conditions	*Pediastrum*, rotifer eggs, filamentous cyanobacteria, *Plumatella* statoblasts	Aquatic habitats; some taxa favoured by nutrient-rich water	Aquatic productivity, eutrophication and changes in trophic state
Grazing/pastoral activity	*Sporormiella*-type	Dung-associated/coprophilous fungi	Herbivore presence and possible grazing pressure
Agricultural/vegetation disturbance	*Thecaphora*, *Tilletia*	Fungi associated with grasses and herbaceous plants	Possible agricultural activity and disturbed vegetation
Marsh/seasonal drying	*Pseudoschizaea* (*Chomotriletes/Moyeria*-type)	Wet soils, freshwater marshes and seasonally dry habitats	Hydrological fluctuations and seasonal moisture stress
Other terrestrial/aquatic remains	Monolete spores, trilete spores, Lepidoptera scales, HdV-989, *Peronospora*-type	Variable ecological affinities	Supporting information; limited use as standalone environmental indicators
Zoological remains	Rotifer-like forms, chironomid/insect fragments, other EIES zoological types	Aquatic or terrestrial invertebrates	Local aquatic and depositional conditions

assemblage, but these are assigned more cautiously as their environmental significance is less specific.

Descriptions of NPPs are given as follows:

Glomus **(**Fig. 4, 1-2): is a genus of mycorrhizal fungi that exhibits various shapes and sizes throughout the core sample. These fungi are ellipsoid, aseptate, and inaperturate, consisting of single cells with slightly thin or smooth, thick walls. Their surfaces range from psilate to very finely scabrate, and most possess hypha-like attachments or appendages, with a maximum size of 62 × 116 μm (Gelorini et al., 2011; Loughlin et al., 2018). *Glomus* forms a vesicular–arbuscular mycorrhizal relationship with flowering plants and occurs on the roots of vegetation (Montoya et al., 2012). The chlamydospores of this genus are dispersed by wind and animals (Camacho et al., 2000). The presence of this genus in the environment is an indicator of soil erosion (Gelorini et al., 2011; van Asperen et al., 2021) (see Fig. 4).

Microthyriaceae **(**Figs. 3 and 4**):** The members of the family Microthyriaceae are epiphyllous (Quamar, 2015). The assemblage of fungal spores, exhibiting great diversity, is recognizable by their non-sporic, easily fossilized structures (Limaye et al., 2007). This fungal family includes common pathogens of herbaceous plants, particularly grasses (Basumatary et al., 2017). Their parasitic forms are usually found in subtropical and tropical regions, while saprotrophic and hyperparasitic forms occur in temperate areas (Hofmann, 2009). Epiphyllous forms, especially on fern foliage, indicate high humidity and heavy rainfall (Limaye et al., 2007; Basumatary et al., 2017).

Ceratophyllum **leaf spinule (**Fig. 4, 4**):** is a submerged, herbaceous, hydrophilous perennial plant without roots. The lineaFr leaves are fused, and their shapes are specialized (dissected). The associated vegetative bud at each node typically forms a zigzag line, likely as an adaptation to the aquatic environment (Iwamoto et al., 2015).

Azolla **(**Fig. 4, 5**):** This sample comprised *Azolla* shoots of various sizes forming a network, with a minimum length of 43 × 143 μm. The shoots are yellow with brown spots. *Azolla* is a freshwater, surface-floating heterosporous fern that bloomed and reproduced in the Arctic Ocean during the early Middle Eocene (van Geel and Aptroot, 2006).

Fungal Hyphae (Fig. 4, 6-9**):** Unicellular to multicellular, dichotomously branched, and composed of chitin, these structures indicate high organic input and heavy rainfall (Limaye et al., 2007).

Spirogyra **(**Fig. 4, 10, 11**):** Spores of *Spirogyra* are oval and radially symmetrical, with a length of 216 μm. *Spirogyra* inhabits clear, oxygen-rich freshwater ponds that fluctuate seasonally, as well as low-water-level environments such as streams and lakes.

Statoblast of *Plumatella* **(**Fig. 4 and 12**):** An oval-shaped asexual reproductive structure in Bryozoa, formed by internal budding and the encasement of the bud within a capsule composed of two tightly fitting, sclerotized, and sculptured valves. Statoblasts are highly resistant structures capable of surviving adverse conditions and germinating to form new colonies when favorable conditions arise. High concentrations of statoblasts in freshwater bryozoans have been documented in organic-rich and polluted waters, including habitats contaminated with livestock waste. Statoblasts are used as indicators to demonstrate the transition from aquatic to lakeshore environments and the increase in riverbank vegetation (Smol et al., 2002; Revelles et al., 2016).

Lepidoptera wing scale (Fig. 4 and 13**):** The order Lepidoptera is the fourth largest insect order in Europe, comprising 8470 species distributed across 35 countries. Due to this species richness, the taxon list of as given area provides an exceptionally detailed view of the environmental conditions at the site. Conversely, butterflies and moths exhibit a sensitive response to changes in abiotic factors. Therefore, butterflies and moths can be considered effective indicator species for monitoring climate change (Hufnagel and Kocsis, 2011).

Filamentous cyanobacteria (Fig. 4 and 14**):** The increase in cyanobacteria reflects excessive eutrophication, providing a competitive advantage over green algae in environments with elevated phosphorus levels (Revelles et al., 2016).

Oospores of *Peronospora* (HdV-819), (Fig. 4, 15, 16**).** "Winged" microfossils feature a central double-walled globose structure measuring approximately 68–81 μm in diameter. This central region is often filled with dark material, frequently containing smaller globose objects. The wings are relatively thin-walled compared to the central structure (Yeloff et al., 2007; NPP database). These structures are globose to sub-globose or irregular in shape, yellow-brown to dark reddish-brown in color, distinctly aplerotic, with walls that are commonly wrinkled and smooth (Voglmayr et al., 2014).

Sporormiella**-type (**Fig. 4, 17-22**):** *Sporormiella* is a dung-coprophilous fungus characterized by separate spore cells (Montoya et al., 2012) or three-to multi-septate ascospores with a sigmoid germinal aperture (Cook et al., 2011). *Sporormiella*-type spores are indicators of grazing activity and/or decaying wood (Mighall et al., 2006; Davis and Shafer, 2006; Loughlin et al., 2018; Lee et al., 2022). *Sporormiella* did not reach high values in ungrazed forests and in densely wooded pastures (Dietre et al., 2012).

HdV-989 (Fig. 4, 23-26**):** Microfossil hyaline, globose, measuring 29–57 μm in diameter, ornamented with numerous densely arranged anastomosing processes. It is similar to HdV-984, except for the presence of anastomosis, which does not occur in HdV-984 (Carrión and Van Geel, 1999; NPP database).

Pseudoschizaea (updated to *Chomotriletes* or *Moyeria*): Microscopic, disc-shaped shells measuring approximately 35 μm, characterized by concentric circular ribs, are abundantly found in various soils and sediment types across different regions of the globe, particularly in freshwater and brackish-water environments (Medeanic, 2006). The synchronous increase of Artemisia and *Pseudoschizaea*, accompanied by a decline in mesophytes after 4500 yr BP, can be interpreted as a consequence of enhanced regional aridity (Carrión and Navarro, 2002). *Pseudoschizaea* cysts are associated with freshwater marshes (Montoya et al., 2012), which occur concurrently with desiccation phases or intensified summer droughts (Carrión and Navarro, 2002; Revelles et al., 2016).

Monolete spore (Fig. 4 and 27**):** These spores are kidney bean-shaped, ranging in color from yellow to brown, and exhibit a smooth, psilate surface texture (Gelorini et al., 2011; Adojoh et al., 2019; Kutluk, 2019). Monolete spores are crucial for understanding the oxygenation of Earth's atmosphere, the emergence of multicellular life in the oceans, and the colonization of land.

Trilete spore (Fig. 4, 28-41**):** The shape is circular to subtriangular. The trilete mark is straight, thin, and extends to the equator. The exine is fairly irregular. The proximal face appears laevigate, while the distal face and equatorial area are ornamented with densely packed, thin bacula.

Rotifera eggs: The phylum Rotifera consists of microscopic metazoans that feed on particulate matter, algae, protozoans, other rotifers, and even small *cladocerans* (McCarthy et al., 2021). They are found in various freshwater environments, moist soils, mosses, and lichens, and can even grow on freshwater crustaceans and aquatic insect larvae across most continents (Cook et al., 2011; McCarthy et al., 2021). Their resting eggs can survive extreme conditions, such as very low temperatures or desiccation. This ability to undergo dormancy enables rotifers to disperse widely (Ricci, 2001).

Pediastrum **(**Fig. 4, 42-43**):** is composed of tightly packed cells, with peripheral cells featuring two short, conical processes flanking a U-shaped concave margin and a cell wall exhibiting distinct, irregular net-like sculpturing. This morphology appears indicative of both large and small lake habitats characterized by slightly alkaline water and abundant submerged macrophyte vegetation.

Scleroderma **(**Fig. 4, 44-45**):** This genus forms ectomycorrhizal associations with several trees and shrubs and is commonly found in sandy soils, humus, grasslands, and occasionally on decaying wood. It may be linked to increased soil erosion following fire events. Indeed, high proportions of spores from ectomycorrhizal and endomycorrhizal fungi (such as *Scleroderma*-type and *Glomus* species) may result from rainwater runoff and/or soil erosion (Dal Corso et al., 2019).



Fig. 4. Light microscopy images of the non-pollen palynomorphs from core JARK014-4. The numbers are as follows: **1, 2** *Glomus* (Glomeromycota); **3** Microthyriaceae; **4** *Ceratophyllum* leaf spinule (HdV 137); **5**- *Azolla*; **6–9** Fungal hyphae; **10**, **11** *Spirogyra*; **12** Statoblast of *Plumatella* (Montoya et al., 2018), **13** Scale of *Lepidoptera* wing; **14**- Filamentous cyanobacteria; **15, 16**- Oospores of *Peronospora* (HdV-819) (Voglmayr et al., 2014); **17-22** *Preussia-Sporormiella-Sporormiella* complex; **23-26** HdV-989/HdV-984; **27**- Monolete spore; **28-41** Trilete spore; **42,43**- *Pediastrum*; **44,-45**- *Scleroderma*; **46**- *Thecaphora;* **47, 48**- *Zygnema;* **49-52** *Tilletia*; (Camacho et al., 2000; Carrión and Navarro, 2002), **53-56** TM-380 or UY-117; UY-109 (Leal et al., 2021).

***Thecaphora* (Fig. 4 and 46):** is a group of fungi frequently parasitic on grasses and may be related to agricultural practices (Carrión and Navarro, 2002; Dal Corso et al., 2019).

***Zygnema* (Fig. 4, 47-48):** is a spheroidal spore with hyaline, pitted surfaces (Leal et al., 2021), typically found in mesotrophic to eutrophic open freshwater environments as green clumps and floating mats. It inhabits shallow eutrophic freshwater habitats (Shumilovskikh et al., 2019) and serves as an indicator of wetter and cooler climates (Simakova et al., 2021). It is commonly found in ponds or wet soil (van Geel, 2001) and produces spores in the spring when conditions are warm.

***Tilletia* (Fig. 4, 49-52):** is ellipsoid or ovoid, with a rudimentary style and stigmas, initially covered by a thin pericarp that is purplish brown, dark brown, or blackish brown. This pericarp later ruptures, exposing a powdery, dark reddish-brown mass of spores and sterile cells. While *Thecaphora* species affect several herbs, *Tilletia* caries parasitizes grasses and cereals, such as Triticum and Secale (Dal Corso et al., 2019).

TM-380 or UY-117; UY-109 (Fig. 4, 53-56): Spherical units with a yellow wall and elongated warts. Similar to UY-109 but echinate and smaller, approximately 16 μm in diameter (Leal et al.).

3.3. Non-pollen palynomorph record

The biodiversity of non-pollen palynomorphs increased from bottom to top within the core. These samples are divided into five zones, from bottom to top (Fig. 5, Fig. 6, and Fig. 7).

3.3.1. Non-pollen zone SE−1 (467–377 cm: 12,000–10,500 cal yrs BP)

Based on our analysis, *Glomus*, with a maximum of 57.46% at 11,250 cal yrs BP, is the dominant non-pollen palynomorph (NPP) in this zone. Its presence is followed by *Azolla*, with 37.66% cal yrs BP, and *Arcella*, with 8.29% at 11,250 cal yrs BP (Fig. 5, Fig. 6)). Other noteworthy NPPs include algae such as *Zygnema* and *Spirogyra*, which show slightly elevated percentages, with 13.5% at 11,500 cal yrs BP. Additionally, spores from the *Diphasium* genus, with a maximum of 8.33% at 10,700 cal yrs BP, and several unidentified monolete fungal spores, with the highest percentage of 5.26% around 11,000 cal yrs BP, increased during this period. The most abundant unknown NPP was categorized as EIES-142, with a maximum of 15.79% at 10,900 cal yrs BP (Fig. 7).

3.3.2. Non-pollen zone SE−2 (377–257 cm: 10,500–7400 cal yrs BP)

This non-pollen zone is divided into two subzones: SE-2a: 377–345 cm; 10,500– 8600 cal yrs BP; and SE-2b: 345–257 cm; 8600–7400 cal yrs BP.

3.3.3. Subzone SE−2a (377–345 cm: 10,500–8600 cal yrs BP)

During this time, the percentage of *Glomus* reached 84.5% at 9740 cal yrs BP (Fig. 5). *Arcella* experienced a steady, stable fluctuation between 1.45% and 3.1%. *Spirogyra* increased to 9.63% at 8600 cal yrs BP, while *Zygnema* decreased dramatically from 14.17% to 0.99% during this period. The spores of *Diphasium*, *Botrychium*, and monolete spores changed steadily, ranging from approximately 0.99% to 5.8%. Some of the new NPPs documented in this phase were: EIES-039, EIES-078, EIES-154, EIES-155, and HdV-10 (Fig. 7).

3.3.4. Subzone SE-2b (345,257 cm: 8600–7400 cal yrs BP)

Glomus dropped abruptly to the minimum percentage of the entire time period. In contrast, *Azolla* peaked at 97.58%, the highest amount recorded in the entire core (Fig. 5 and Fig. 6). Sharp fluctuations were observed in *Spirogyra* percentages, ranging from 0.23% to 29.81%. Additionally, several coded nomenclatures such as EIES-041, EIES-044, EIES-094, EIES-097, EIES-119, EIES-157, and EIES-162 were recorded as new species in this zone (Fig. 7).

3.3.5. Non-pollen zone SE−3 (257–203 cm: 7400–5000 cal yrs BP)

This zone is divided into two subzones: SE-3a: 257–237 cm; 7400–6600 cal yrs BP; SE-3b: 237–203 cm; and 6600– 5000 cal yrs BP.

3.3.6. Subzone SE−3a (257–237 cm: 7400– 6600 cal yrs BP)

This subzone is characterized by a sharp increase in *Glomus* (maximum 75.82%), *Sporormiella* type (maximum 0.73%), and *Azolla* (maximum 5.839%) (Fig. 5). Monolete spores, *Botrychium*, *Diphasium*, and Trilete spores (ranging from 0.64% to 2.92%) exhibit a steady, stable trend during this period. EIES-035, EIES-124, and EIES-158 are some of the new species documented in this zone (Fig. 7).

3.3.7. Subzone SE-3b (237–203 cm: 6600–5000 cal yrs BP)

The most abundant fungus is *Glomus* (87.77%), followed by *Tilletia* (2.43%) and *Thecaphora* (7.98%). *Arcella* increased from 1.57% to 9.97% during this period. *Spirogyra* and *Zygnema* showed slight

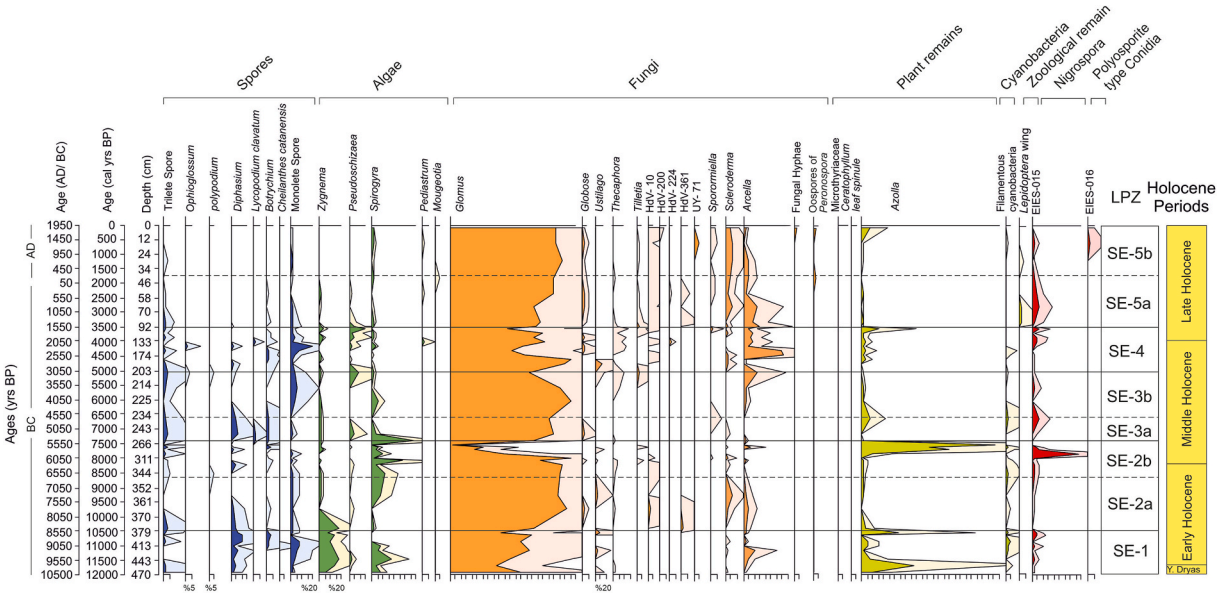


Fig. 5. The detailed non-pollen palynomorph diagram shows the non-pollen palynomorph percentages in the JARK14−04 sediment core (exaggeration, 10x). The identification of non-palynomorph zones was carried out through the application of CONISS within the Tilia program described by Grimm (1987), and the diagram divided into five non–pollen zones (SE1− SE5). LPZ is the abbreviation of local non pollen zones.

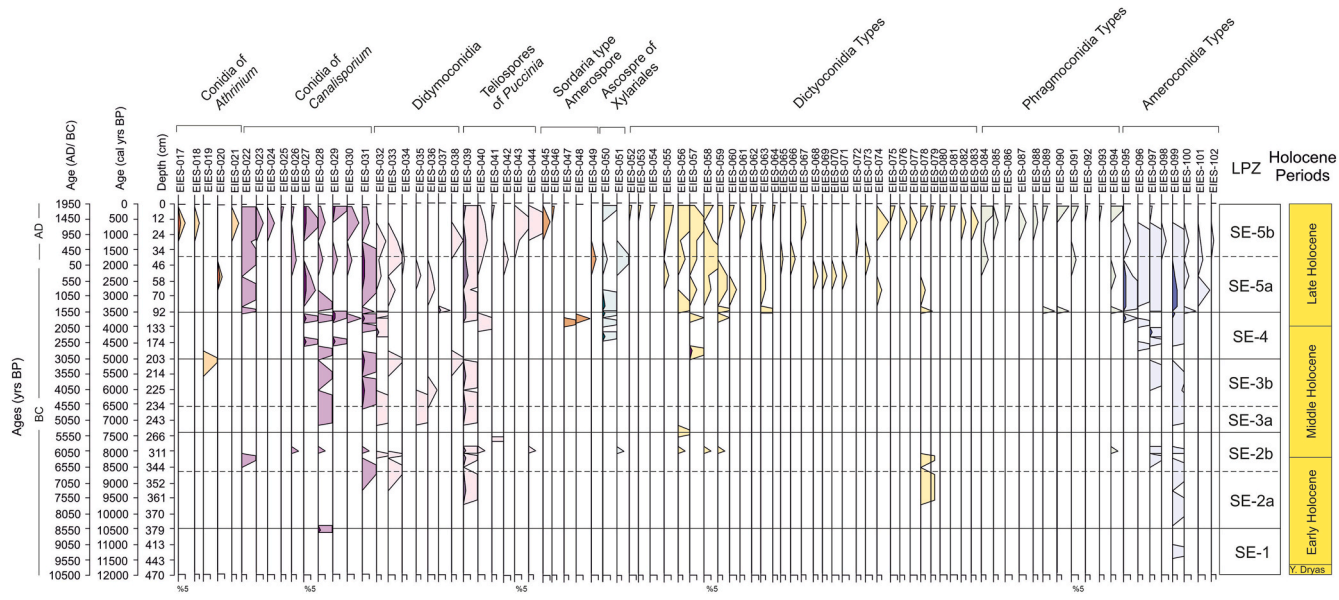


Fig. 6. The continuation of the detailed non-pollen palynomorph diagram shows the non-pollen palynomorph percentages in the JARK14−04 sediment core (exaggeration, 10x). LPZ is the abbreviation of local pollen zones.

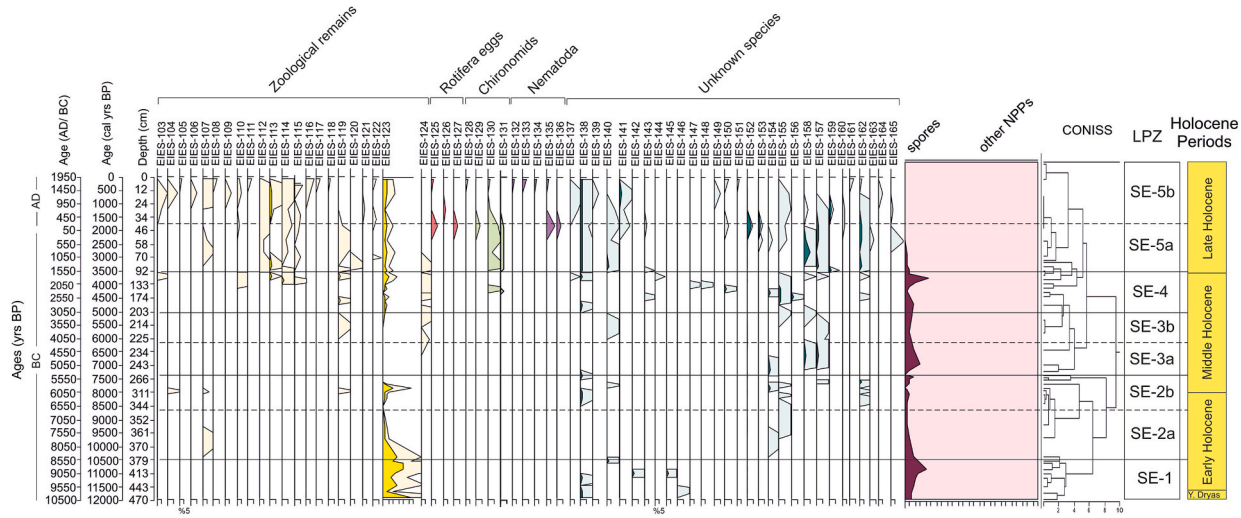


Fig. 7. The continuation of the detailed non-pollen palynomorph diagram shows the non-pollen palynomorph percentages in the JARK14−04 sediment core (exaggeration, 10x). LPZ is the abbreviation of local pollen zones.

fluctuations. Monolete spores, Trilete spores, and *Botrychium* exhibit a relatively stable trend around 2%. EIES-019, EIES-036, and EIES-038 emerged during this time (Fig. 7).

3.3.8. Non-pollen zone SE−4 (203−91 cm: 5000− 3500 cal yrs BP)

This zone is characterized by intense fluctuating in *Glomus* (41.5% and 88.99%), *Thecaphora* (0.39% and 11.32%), *Arcella (0.22% and 13.2%)*, and *Azolla* (0.47% and 28.85%) (Fig. 5). The slight fluctuations in *Spirogyra* (0.38% and 2.09%), *Chromotriletes* (*Pseudoschizaea)* (0.45% and 11.9%), *Zygnema* (0.36% and 3.1%), *Botrychium (0.35% and 0.95%)*, and Trilete spore (0.36% anf 2.4%) happened. Some new nomenclatures such as EIES-027, EIES-029, EIES-048, EIES-050, EIES-057, EIES-096, EIES-130, EIES-148, and EIES-150 exists in this zone for the first time.

3.3.9. Non-pollen zone SE−5 (91–1 cm: 3500− subrecent)

This zone is studied in two parts.

3.3.10. Subzone SE−5a (91–41 cm: 3500–1750 cal yrs BP)

The prominent taxa of the core, *Glomus* (min: 63.1% and max: 76.2%), fluctuated slightly during this time period. *Arcella* (3.1% to 11.9%) increased, and *Scleroderma* (1.6% and 4.7%) fluctuated. *Botrychium* fluctuated in low percentages less than 1%, and monolete spore fluctuated highly from 0.065% to 1.7%. EIES-056, EIES-063, EIES-068, EIES-074, EIES-070, EIES-069, EIES,073, EIES-071, EIES-160, EIES-107, EIES-112, and EIES-152 emerged during this time period (Fig. 7).

3.3.11. Subzone SE−5b (41–1 cm: 1750 cal yrs BP to present)

Glomus high percentage continues in this zone with slight fluctuations between 76.2% and 81.4%. *Scleroderma* (between 3% and 5%), *Arcella* (between 0.9% and 3.4%), *Azolla* (0.049% and 6.1%), *Spirogyra* (between 0.31% and 1.5%), and Monolete spores (0.09% and 0.4%) exists in low amounts. HdV-200, EIES-017, EIES-018, EIES-021, EIES-023, EIES-024, EIES-025, EIES-043, EIES-052, EIES-053, EIES-054, EIES-079, EIES-087, EIES-088, EIES-092, and EIES-093 were

documented in this subzone.

4. Discussion

This study presents the first detailed reconstruction of paleoenvironmental changes at Seki-Eceler Höyük based on NPP over the last 12,000 yr. Changes in NPP assemblages provide evidence of variations in hydrological conditions, erosion, aquatic environments and human activities. However, possible fire-related signals are interpreted cautiously, as independent charcoal-based fire history data were not evaluated in this study.

4.1. Paleoenvironmental dynamics

The early Holocene NPP record shows that relatively extensive freshwater and wetland habitats persisted in the Seki Basin. High concentrations of *Zygnema*, *Azolla*, and *Spirogyra*, along with the presence of *Arcella*, are robust indicators of a thriving, well-watered environment. The stratigraphic presence of monolete and trilete fungal spores corroborates this interpretation, as these taxa are typically found in well-oxygenated riparian zones and healthy soil. The fluctuation of *Glomus* suggests that the region was stable, with limited erosion and infrequent, low-intensity fire events. These findings are consistent with the results of previous pollen studies in the area, which documented humidity oscillations during the early Holocene. Notably, the absence of fungal spores such as *Tilletia*, *Sporormiella* type, and *Thecaphora* indicate that the early Holocene NPP assemblage contains little evidence of intensive human disturbance. In particular, *Sporormiella* type and other NPPs that are potentially associated with herbivore presence or disturbed vegetation either do not occur or occur at very low abundances. It is important to note that *Tilletia* and *Thecaphora* are not considered direct indicators of pastoralism here; these fungi are mainly associated with grasses and herbaceous hosts, and they can only provide supporting evidence of vegetation disturbance or agricultural activity when examined alongside other independent indicators.

The early Holocene period saw a thriving, well-watered environment, as evidenced by our findings, which align with other contemporaneous records. For instance, studies of Black Sea sediments from this period have revealed *Gloeotrichia*-type sheaths and the jaws of ostracods of an unknown species, indicating freshwater environments (Mudie et al., 2011). Research on Lake Hazar indicates that the lake level was approximately 70–80 m lower during the early Holocene than it is today, reinforcing the regional pattern of climatic and hydrological changes at the beginning of this epoch (Öntürk and Kutluk, 2022). The presence of non–pollen palynomorph indicators of fire, soil erosion, or human settlement during the early Holocene is demonstrated by our data, aligning with other studies from the period. However, some researchers have documented early signs of human or megafaunal impact. For instance, an increase in *Sporormiella* type between 10,750 and 10,200 yr BP in some records has been linked to the incursion of megaherbivores such as bison (Mudie et al., 2011). We found that there were no livestock-related fungi in the early Holocene, which contrasts with our findings in the later Neolithic period. Other NPP studies have successfully documented the environmental impacts of early agriculture and pastoralism, such as deforestation of the landscape at Lake Banyoles in Spain (Revelles et al., 2016) and increasing eutrophication due to human livestock activities (Hillbrand et al., 2014).

The period between 8200 and 7700 yr BP is marked by clear indicators of increased aridity, erosion, and deforestation. The sudden disappearance of *Azolla*, *Zygnema*, and *Spirogyra*, which are aquatic and wetland indicators, points to a significant reduction in surface water (Fig. 5). This is coupled with a gradual increase in *Glomus* and *Scleroderma*, which are typically associated with increased erosion and arid conditions, respectively (Fig. 5). Together, these changes may reflect reduced effective moisture and increased disturbance to the catchment area associated with the 8.2 ka climatic event. However, the NPP assemblage alone cannot definitively distinguish between climatic and hydrological or catchment processes. This is also suggested by a decline in arboreal pollen and the presence of arid-indicator species such as *Centaurea solstitialis*, Apiaceae and *Artemisia*, confirming a shift to a cold, arid climate (Hashemi et al., 2026a,b). This interpretation is broadly consistent with Holocene hydroclimatic and vegetation records from Anatolia. Nearby, at Lake Gölhisar in southwestern Türkiye, pollen records indicate that the landscape was initially relatively open, followed by a marked expansion of arboreal vegetation during the early Holocene. This eventually resulted in mixed forests dominated by *Quercus*, *Pinus* and *Juniperus* (Eastwood et al., 1999). Further east, the Lake Hazar pollen record shows that deciduous *Quercus* expanded in the early Holocene, which is consistent with increasing humidity and afforestation (Biltekin et al., 2018). The pollen, isotope and geochemical records from Lake Van also indicate an increase in moisture at the beginning of the Holocene, following the cold, dry Younger Dryas period (Wick et al., 2003). Together, these records suggest that changes to the Seki-Eceler NPP occurred within a broader pattern of significant Holocene hydroclimatic and vegetation variability across Anatolia. However, the magnitude and timing of these responses varied between different sites.

For example, the $\delta^{18}O$ record from Kocain Cave clearly shows the 8.2 ka event as a significant shift towards drier conditions (Jacobson et al., 2021). This broadly corresponds in timing with the reduction in aquatic NPPs and the increase in fungal indicators related to erosion and dry conditions observed at Seki-Eceler. This regional hydroclimatic record, which is independent of the Seki-Eceler data, supports our cautious interpretation that the changes in the Seki-Eceler NPP assemblage between 8200 and 7700 yr BP may be associated with the 8.2 ka event rather than being solely the result of catchment processes. The agreement with the Kocain Cave record also situates the NPP-based observations within a southwestern Anatolian hydroclimatic context, providing an additional line of evidence for the inferred environmental change.

Beyond the 8.2 ka event, changes in aquatic and terrestrial NPP assemblages at Seki-Eceler suggest ongoing hydrological and catchment variability during the middle Holocene. These changes can be compared with hydroclimatic oscillations on centennial to multi-centennial scales documented in Anatolian speleothem records. For instance, the $\delta^{18}O$ record from Dim Cave reveals a swift shift from wetter to drier conditions between around 4320 and 3870 yr BP (Baykara et al., 2026). The occurrence of comparable hydroclimatic variability in an independent regional archive suggests that at least some of the variation recorded by the Seki-Eceler NPP assemblages may reflect broader regional hydroclimatic changes.

Following this arid phase, the period between 7700 and 7500 yr BP is characterized by an abrupt environmental reversal. The sudden increase in *Azolla* and *Spirogyra* indicates the rapid return of freshwater to the region. At the same time, *Glomus* and *Scleroderma* concentrations dropped to their lowest levels, suggesting a decline in erosion and an expansion of forested landscapes (Fig. 5). This shift marks the beginning of the Middle Holocene Climatic Optimum (MCO) in the Seki region, a period characterized by increased humidity and more favorable climatic conditions. Following this brief period of increased freshwater input, the sudden increase in *Glomus*, alongside a corresponding decrease in freshwater indicators such as *Spirogyra*, *Arcella* and *Zygnema*, indicates a significant shift towards reduced moisture availability and increased catchment disturbance. This pattern occurred against the backdrop of vegetation development and hydroclimatic variability in Anatolia during the middle Holocene. At Lake Gölhisar, mixed woodland remained well developed for much of this period (Eastwood et al., 1999). The presence of deciduous *Quercus* populations at Lake Hazar also suggests generally favorable conditions for woodland development during the middle Holocene (Biltekin et al., 2018).

As summarised in Table 2, the ecological classification of the main NPP groups indicates that the paleoenvironmental interpretation of the Seki-Eceler record is primarily based on taxa with established ecological

affinities and assemblage-level changes, rather than individual NPP occurrences. Within this context, the *Sporormiella* type is classified as a coprophilous fungal indicator associated with the presence of herbivores and potential grazing pressure (Table 2).

The presence of *Sporormiella* type during the middle Holocene suggests the presence of herbivores, but it cannot be considered direct evidence of pastoralism or livestock husbandry. Although *Sporormiella* type is commonly used as a coprophilous fungal indicator in archaeological and palaeoecological studies, its spores may originate from the dung of both domestic livestock and wild herbivores (Davis and Shafer, 2006; Raper and Bush, 2009). Therefore, an increase in *Sporormiella* type may indicate enhanced herbivore presence or grazing pressure, but distinguishing anthropogenic grazing from the activity of wild herbivores requires independent supporting evidence, such as archaeological data and pollen indicators of vegetation disturbance and land use. Consequently, the occurrence of *Sporormiella* type at Seki-Eceler is interpreted as evidence of herbivore activity and possible grazing rather than as direct evidence of livestock husbandry. Where this signal coincides with independent evidence of vegetation opening or anthropogenic disturbance, a contribution from human land use may be considered, but cannot be demonstrated from the NPP record alone. The abundance and representation of *Sporormiella* type may also be influenced by depositional and taphonomic processes, including distance from grazing areas and the depositional site, transport, hydrological conditions, and sediment focusing (Raper and Bush, 2009; Raczka et al., 2016). These factors further emphasize the need to interpret *Sporormiella* type within a multiproxy and site-specific framework rather than as a standalone indicator of pastoral activity. The paleoenvironmental record from 5500 to 5000 yr BP reveals a significant shift towards agricultural practices, marked by the increased presence of certain non-pollen palynomorphs (NPPs). The sharp decline of *Glomus*, which is often associated with soil erosion in natural landscapes. *Thecaphora* and *Tilletia* are associated with grasses and herbaceous hosts. Their presence may therefore indicate the disturbance of herbaceous vegetation and potentially agricultural activity, provided this is supported by other evidence. The pollen record from the same core is important in this respect. An increase in Poaceae and other indicators of vegetation disturbance provides additional evidence of increasing human modification of the landscape (Hashemi et al., 2026a,b). Therefore, the interpretation of agricultural activity during this period relies on the correlation between NPP and pollen evidence rather than on *Thecaphora* or *Tilletia* alone. Despite the evidence of agricultural expansion, the presence of *Spirogyra* and *Chromotriletes* indicates that the climate remained relatively humid. *Chromotriletes* is an important indicator species because it is found in warm terrestrial environments with strong seasonal moisture fluctuations. It often thrives in damp soils, such as those found in springs, swamps, and alluvial plains (Scott, 1992; Shumilovskikh et al., 2019). The co-occurrence of these moisture indicators alongside agricultural proxies suggests that, although human activities were altering the landscape, the regional climate still provided enough water for cultivation and the persistence of some wetlands. This pattern of intensified human-induced soil disturbance during this period is not unique to Seki-Eceler Höyük. The late Holocene period at Seki-Eceler Höyük is characterized by a significant decline in aquatic environments and a concurrent increase in erosion. The near disappearance of freshwater algae such as *Spirogyra* and *Zygnema*, as well as the aquatic fern *Azolla*, suggests a substantial decrease in the amount of available surface water. Concurrently, an increase in *Glomus* and *Scleroderma* suggests intensified erosion and drier soil conditions. Notably, this period also exhibits a decrease in indicators of husbandry and agricultural activities, suggesting either a change in human land use or a response to evolving environmental conditions.

Indeed, the broader paleoecological literature from southwestern Anatolia indicates increasing rather than decreasing anthropogenic pressure during much of the late Holocene. Evidence from Lake Gölhisar in southwestern Türkiye, close to Seki-Eceler, is particularly relevant. Pollen records from this location indicate that a mixed *Quercus–Pinus–Juniperus* woodland persisted until around 3000 yr BP, after which a significant period of human-driven landscape transformation emerged (Eastwood et al., 1999). This period was characterised by deforestation, cereal cultivation, pastoralism and an increased presence of cultivated trees, including *Olea*, *Juglans*, *Castanea* and *Vitis*. Changes in aquatic ecology were also recorded after 3000 yr BP, which were related to increased sediment and nutrient transfer from the catchment area. Therefore, the increase in disturbance-related NPPs and the reduction in aquatic indicators at Seki-Eceler may have occurred within a broader phase of intensified landscape modification in southwestern Anatolia. A comparable record of vegetation disturbance and agricultural expansion has been documented in the Sagalassos region of the Western Taurus Mountains in Türkiye. At Gravgaz, the late Holocene pollen record indicates an initially pine-dominated landscape around 3300–3200 yr BP, followed by a prolonged deforestation phase around 3200–2300 yr BP and a major cultivation phase around 2300–1300 yr BP (Vermoere et al., 2000). 3300–3200 yr BP, followed by prolonged deforestation between 3200 and 2300 yr BP, and major cultivation between 2300 and 1300 yr BP (Vermoere et al., 2000). Additional record from Gravgaz suggest that human activity became more apparent from around 2530 yr BP, with deforestation and landscape disturbance preceding the intensive cultivation of olive trees (Vermoere et al., 2002). Although the Beyşehir Occupation Phase (BOP) is expressed weakly at Seki-Eceler (Hashemi et al., 2026a,b), these transformations broadly correspond to the Beyşehir Occupation Phase (BOP), a major episode of intensified land use recognised in numerous Anatolian pollen records.

Direct comparisons between Seki-Eceler, Lake Azaplı and Lake Sülük are particularly informative, as these sites exhibit different responses in non–pollen palynomorphs (NPP) to late Holocene climatic and anthropogenic influences. At Lake Azaplı in south-eastern Türkiye, evidence from pollen, NPP and charcoal indicates substantial changes in catchment conditions and increasing human influence during the mid-to late Holocene. The occurrence of cultivated trees, including *Juglans*, *Pistacia*, and *Olea europaea*, together with agropastoral NPP indicators, suggests intensified human activity between approximately 3250 and 2520 yr BP (Gökgöz et al., 2026). The Azaplı record therefore primarily emphasises increasing agropastoral activity and associated catchment disturbance. In contrast, the late Holocene NPP record from Lake Sülük is more strongly characterized by changes within the aquatic ecosystem. Increased rotifer eggs suggest enhanced eutrophication, while the abundance of *Azolla* indicates the persistence or expansion of shallow, nutrient-rich freshwater habitats (Biltekin et al., 2025). At Seki-Eceler, however, the response is different: the decline or disappearance of *Azolla*, *Spirogyra* and *Zygnema*, together with an increase in *Glomus* and *Scleroderma*, indicates a contraction of freshwater habitats, accompanied by an increase in catchment disturbance. Therefore, while the Lake Azaplı record predominantly captures an agropastoral/disturbance signal, the Lake Sülük record captures an aquatic/eutrophication signal, and the Seki-Eceler record is characterized by declining freshwater availability and increasing catchment instability. These contrasting NPP trajectories are important because they show that the environmental changes that occurred during the late Holocene did not produce a uniform palynological response across Anatolia. Instead, broader climatic and anthropogenic influences were strongly mediated by basin morphology, hydrological regimes, nutrient availability, catchment characteristics and land use intensity. The contrast between the abundance of *Azolla* at Lake Sülük and its decline at Seki-Eceler provides a clear example of this: the same broad late Holocene interval supported the expansion of shallow, nutrient-rich aquatic habitats in one basin, while in another, freshwater habitats contracted. The agropastoral signal at Lake Azaplı shows that human activity can be primarily expressed through fungal and vegetation indicators rather than through changes to the aquatic ecosystem. In Fore-Baikal, Russia, the presence of freshwater algae indicates periods of low water flow (Khenzykhenova et al., 2019). The presence of *Pediastrum* in the Black Sea corridor

indicates eutrophic, hard-water lakes (van Geel, 2001; cited in Mudie et al., 2010). Studies of rotifer eggs in Lake Sülük also showed periodic increases in eutrophication (Biltekin et al., 2025). In the Ligurian Apennines in Italy, coprophilous fungi provided evidence of the presence of livestock (Menozzi et al., 2010). Likewise, the presence of parasitic worms in Lake Hazar suggests that agropastoral activities have been present since around AD 500 (Öntürk and Kutluk, 2022). Tunno and Mensing (2017) documented a dry period, while Srivastava et al. (2013) used *Botryococcus*, *Pediastrum*, and *Arcella* to track hydrological changes in Pichavaram, India. However, it is worth noting that these algae may have different elevation tolerances; for example, *Pediastrum* has been found at higher altitudes than *Botryococcus* (Rull et al., 2008). These varied records from different geographical locations highlight the importance of interpreting NPP assemblages to understand the paleoenvironmental history of a given site.

Further east, the Lake Hazar record provides independent, multiproxy evidence of the complexity of late Holocene climatic and anthropogenic interactions. Between 3.2 and 2.9 ka BP, higher lake levels, increased river discharge, higher arboreal pollen (particularly from deciduous *Quercus*) and changes in Ca/Ti, total organic carbon (TOC) and grain size indicate relatively wetter conditions (Biltekin et al., 2021). However, the longer Holocene pollen record from Lake Hazar documents increased herbaceous vegetation and reduced lake levels during other periods of the late Holocene, indicating intervals of colder, drier conditions (Biltekin et al., 2018). These alternating conditions demonstrate that the late Holocene in Anatolia cannot be characterized as simply a period of progressive aridification. A high-resolution record from Lake Akgöl in northern Türkiye provides further evidence of this interaction during the latest Holocene. Pollen-based climate reconstructions indicate an increase in plant-available moisture towards approximately 800 yr BP, followed by a shift towards drier conditions. In addition, anthropogenic indicators, including *Plantago*, *Rumex*, *Cerealia*-type pollen and increased non-arboreal pollen, document progressive landscape disturbance and agricultural expansion, particularly during the last ca. 500 years (Biltekin et al., 2026). The findings from Seki-Eceler Höyük offer a distinctive insight into the late Holocene in southwestern Türkiye, a region that had previously lacked detailed NPP data. They also contribute to the wider discussion about human and climatic influences during this period.

4.2. New documented species interpretation

In total, we counted 173 NPPs in core JARK014-4. Of these, 23 were identified and 150 were unidentified and coded as EIES for presentation in this article. The concentration of NPPs in the top of the core was highest and decreased towards the bottom. However, the abundance of fungi indicates that the forestation rate was low in this region. The abundance of algae towards the bottom of the core indicates high humidity during the early Holocene compared to the late Holocene (Figs. 5, 6 , and 7). The ecological significance of the newly documented EIES-coded morphotypes is uncertain because many cannot yet be confidently assigned to modern taxa. Their primary value therefore lies in documenting assemblage diversity and providing a morphological reference for future comparative studies. Until their modern affinities and ecological requirements are more clearly defined, these morphotypes should not be used as standalone indicators of specific climatic, hydrological or anthropogenic processes.

5. Conclusions

Investigating the JARK014–4 core assemblage provides a detailed, NPP–based reconstruction of environmental changes in the Seki-Eceler in the southwestern Türkiye over the last 12,000 years. During the Younger Dryas–early Holocene transition, the abundance of NPPs related to freshwater, including *Azolla*, *Spirogyra*, and *Zygnema*, indicates the presence of relatively extensive aquatic and wetland habitats. However, these assemblages primarily reflect hydrological conditions and should not be interpreted as direct measures of regional humidity. Between 8200 and 7700 yr BP, the NPP record shows a marked decline in aquatic indicators, accompanied by an increase in *Glomus* and *Scleroderma*. This is consistent with reduced freshwater availability and enhanced catchment disturbance. The broad chronological correspondence of these changes with independent regional hydroclimatic records suggests a possible response to the 8.2 ka climatic event, although hydrological and catchment processes cannot be ruled out. Following this period, the significant increase in *Azolla*, *Spirogyra* and *Zygnema*, alongside the decline in *Glomus*, suggests the expansion of freshwater habitats and reduced catchment disturbance. Rather than attributing this change directly to the Middle Holocene Climatic Optimum, it is more cautious to interpret the assemblage as reflecting a phase of relatively favorable hydrological conditions within a period of broader Holocene hydroclimatic variability. From around 3500 yr BP onwards, the decline of NPPs related to freshwater, together with the relatively high abundance of *Glomus* and *Scleroderma*, indicates the increased disturbance of catchments. These changes occurred during a period of increasing anthropogenic pressure, as documented in southwestern Anatolian palaeoecological records. However, the Seki-Eceler NPP record alone does not allow the relative contributions of hydroclimatic variability and human land use to be quantified. The JARK014-4 core contains a highly diverse assemblage of identified, tentatively assigned and taxonomically unresolved NPP morphotypes. The paleoenvironmental interpretations in this study are based primarily on taxa with established ecological significance. The EIES-coded unidentified and tentatively assigned morphotypes are presented principally as a morphological reference and are not used as standalone environmental indicators. The value of these morphotypes for paleoenvironmental reconstruction could be substantially improved by future taxonomic identification and ecological calibration.

CRediT authorship contribution statement

Zahra Hashemi: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Demet Biltekin:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Alan M. Greaves:** Writing – review & editing, Resources, Project administration. **Sena Akçer-Ön:** Writing – review & editing, Visualization, Resources, Formal analysis, Data curation. **Zeki Bora Ön:** Methodology, Formal analysis, Data curation. **Memet Namık Çağatay:** Writing – review & editing, Methodology. **Ömer Yetemen:** Writing – review & editing.

Declaration of competing interest

There are no conflicts of interest associated with the results and/or discussion reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.quaint.2026.110488.

References

Adojoh, O., Fabienne, M., Duller, R., Osterloff, P., 2019. Taxonomy and phytoecology of palynomorphs and non-pollen palynomorphs: a refined compendium from the West Africa Margin. Biodivers. Int. J. 3 (5), 188–200. https://doi.org/10.15406/bij.2019.03.00145.

Aksoy, B., Köse, O., 2005. A site in the Seki Plateau (Lycia): Eceler Höyük. Anatol. antiq. 13 (1), 71–83. https://doi.org/10.3406/anata.2005.1038.

Alçiçek, H., Wesselingh, F.P., Alçiçek, M.C., 2015. Paleoenvironmental evolution of the late Pliocene-early Pleistocene fluvio-deltaic sequence of the Denizli Basin (SW Turkey). Palaeogeogr. Palaeoclimatol. Palaeoecol. 437, 98–116. https://doi.org/10.1016/j.palaeo.2015.06.019.

Bakker, J., Paulissen, E., Kaniewski, D., De Laet, V., Verstraeten, G., Waelkens, M., 2012. Man, vegetation and climate during the Holocene in the territory of Sagalassos, Western Taurus Mountains, SW Turkey. Veg. Hist. Archaeobotany 21, 249–266. https://doi.org/10.1007/s00334-011-0312-4.

Basumatary, S.K., McDonald, H.G., Gogoi, R., 2017. Pollen and non-pollen palynomorph preservation in the dung of the Greater One-horned Rhino (Rhinoceros unicornis), and its implication to palaeoecology and palaeodietary analysis: a case study from India. Rev. Palaeobot. Palynol. 244, 153–162. https://doi.org/10.1016/j.revpalbo.2017.05.004. ISSN 0034-6667.

Baykara, M.O., Özkul, M., Shen, C., İmer, E.Ü., Mii, H., Hu, H., Wu, C., Kele, S., Dem'eny, A., 2026. Mid-to-late Holocene climatic changes in the Eastern Mediterranean region: speleothem-based high-resolution isotope record from Dim Cave, southern Türkiye. J. Quat. Sci. 41 (1), 61–79. https://doi.org/10.1002/jqs.70027.

Biltekin, D., Eriş, K., Çağatay, N., Akçer Ön, S., Bal Akkoca, D., 2018. Late Pleistocene-Holocene environmental change in eastern Turkey: multiproxy palaeoecological data of vegetation and lake-catchment changes. J. Quat. Sci. 33 (5), 575–585.

Biltekin, D., Eriş, K.K., Bulut, S., 2021. Anthropogenic influences and climate changes in Lake Hazar (eastern Turkey) during the Late Holocene. Quat. Int. 583, 70–82. https://doi.org/10.1016/j.quaint.2021.02.023.

Biltekin, D., Eriş, K.K., Schachner, A., Yakupoğlu, N., Yakupoğlu, C., 2025. Late Holocene vegetation dynamics and climate variations through pollen analysis of sediments from Lake Sülük (Çorum, Türkiye). Rev. Palaeobot. Palynol. 334, 105280. https://doi.org/10.1016/j.revpalbo.2025.105280.

Biltekin, D., Oy, H., Tunca, E., Haghipour, N., Acar, İ., Kolbüken, M., Acar, D., Mohammadi, A., 2026. Late Holocene climate and vegetation changes at Lake Akgöl (Samsun, Türkiye). Rev. Palaeobot. Palynol. 351, 105587. https://doi.org/10.1016/j.revpalbo.2026.105587.

Blaauw, M., Christen, J.A., 2011. Flexible paleoclimate age-depth models using an autoregressive gamma process. Bayesian Anal. 6, 457–474. https://doi.org/10.1214/11-BA618.

Camacho, C.N., Carrión, J.S., Navarro, J., Munuera, M., Prieto, A.R., 2000. An experimental approach to the palynology of cave deposits. J. Quat. Sci.: Publ. Quatern. Res. Asso. 15 (6), 603–619. https://doi.org/10.1002/1099-1417(200009)15:6<603:AID-JQS544>3.0.CO;2-F.

Carrión, J.S., Navarro, C., 2002. Cryptogam spores and other non-pollen microfossils as sources of palaeoecological information: case-studies from Spain. Ann. Bot. Fenn. 39 (1), 1–14. http://www.jstor.org/stable/23726867.

Carrión, J.S., Van Geel, B., 1999. Fine-resolution Upper Weichselian and Holocene palynological record from Navarrés (Valencia, Spain) and a discussion about factors of Mediterranean forest succession. Rev. Palaeobot. Palynol. 106 (3–4), 209–236.

Cenk, S., Turgay, P., 2020. The effects of Mediterranean oscillation on temperature and precipitation data in Turkey. JWCC apc 11 (3), 722–743. https://doi.org/10.2166/wcc.2019.192.

Collins, A.S., Robertson, A.H.F., 1998. Processes of Late Cretaceous to late Miocene episodic thrust-sheet translation in the Lycian Taurides, SW Turkey. J. Geol. Soc. 155 (5), 759–772. https://doi.org/10.1144/gsjgs.155.5.0759.

Cook, E.J., Van Geel, B., Van Der Kaars, S., Van Arkel, J., 2011. A review of the use of non-pollen palynomorphs in palaeoecology with examples from Australia. Palynology 35 (2), 155–178. https://doi.org/10.1080/01916122.2010.545515.

Core Team, 2024. R: a Language and Environment for Statistical Computing. R foundation for statistical computing, Vienna, Austria. https://www.R-project.org.

Cour, P., 1974. Nouvelles techniques de détection des flux et des retombées polliniques: étude de la sédimentation des pollens et des spores à la surface du sol. Pollen Spores 16, 103–141.

Criado-Aldeanueva, F., Soto-Navarro, J., 2020. Climatic indices over the Mediterranean Sea: a review. Appl. Sci. 10 (17), 5790. https://doi.org/10.3390/app10175790.

Cugny, C., 2011. Apports Des Microfossiles Non-polliniques À L'Histoire Du Pastoralisme Sur Le Versant Nord Pyrénéen Entre Référentiels Actuels Et Reconstitution Du Passé. PhD Thesis. Toulouse University, p. 261.

Cugny, C., Mazier, F., Galop, D., 2010. Modern and fossil non-pollen palynomorphs from the Basque mountains (western Pyrenees, France): the use of coprophilous fungi to reconstruct pastoral activity. Veg. Hist. Archaeobotany 19, 391–408. https://doi.org/10.1007/s00334-010-0242-6.

Dal Corso, M., Kirleis, W., Kneisel, J., Taylor, N., Wieckowska-Lüth, M., Zanon, M., 2019. How's Life? Living Conditions in the 2nd and 1st Millennia BCE. Sidestone Press, p. 220.

Danelian, T., Robertson, A.H.F., Collins, A.S., Poisson, A., 2006. Biochronology of Jurassic and Early Cretaceous radiolarites from the Lycian Mélange (SW Turkey) and implications for the evolution of the Northern Neotethyan ocean. In: Robertson, A.H.F., Mountrakis, D. (Eds.), Tectonic Development of the Eastern Mediterranean Region, vol. 260. Geological Society of, London, pp. 1–9. https://doi.org/10.1144/GSL.SP.2006.260.01.10.

Davis, O.K., Shafer, D.S., 2006. *Sporormiella* fungal spores, a palynological means of detecting herbivore density. Palaeogeogr. Palaeoclimatol. Palaeoecol. 237 (1), 40–50. https://doi.org/10.1016/j.palaeo.2005.11.028.

Dietre, B., Gauthier, É., Gillet, F., 2012. Modern pollen rain and fungal spore assemblages from pasture woodlands around Lake Saint-Point (France). Rev. Palaeobot. Palynol. 186, 69–89. https://doi.org/10.1016/j.revpalbo.2012.07.002.

Eastwood, W.J., Roberts, N., Lamb, H.F., Tibby, J.C., 1999. Holocene environmental change in southwest Turkey: a palaeoecological record of lake and catchment-related changes. Quat. Sci. Rev. 18, 671–695. https://doi.org/10.1016/S0277-3791(98)00104-8.

Ellis, M.B., Ellis, P., 1997. Microfungi on Land Plants. Richmond Publishing, Slough.

Ellis-Adam, A.C., van Geel, B., 1978. Fossil zygospores of Debarya glyptosperma (De Bary) Wittr. (Zygnemataceae) in Holocene sandy soils. Acta Bot. Neerl. 27, 389–396. https://doi.org/10.1111/j.1438-8677.1978.tb00308.x.

Gelorini, V., Verbeken, A., van Geel, B., Cocquyt, C., Verschuren, D., 2011. Modern non-pollen palynomorphs from East African lake sediments. Rev. Palaeobot. Palynol. 164, 143–173. https://doi.org/10.1016/j.revpalbo.2010.12.002.

Gökgöz, D.D., Biltekin, D., Eriş, K.K., Uçarkuş, G., Kırkan, E., Yakupoğlu, C., Acar, D., Yetemen, Ö., 2026. Mid- to late holocene palaeoenvironmental and palaeoecological dynamics in Lake Azaplı (Gölbaşı Basin, SE Türkiye): insights from pollen and non-pollen palynomorph records. Palaeogeogr. Palaeoclimatol. Palaeoecol. 700, 114085. https://doi.org/10.1016/j.palaeo.2026.114085.

Grimm, E.C., 1987. CONISS: a Fortran 77 program for stratigraphically constrained cluster analysis by the method of incremental sum of squares. Comput. Geosci. 13, 13–35. https://doi.org/10.1016/0098-3004(87)90022-7.

Grimm, E.C., 1991-2015. Tilia Version 2.0.41. Illinois State Museum, Springfield, Illinois, USA [software].

Gurdebeke, P.R., Pospelova, V., Mertens, K.N., Dallimore, A., Chana, J., Louwye, S., 2018. Diversity and distribution of dinoflagellate cysts in surface sediments from fjords of western Vancouver Island (British Columbia, Canada). Mar. Micropaleontol. 143, 12–29. https://doi.org/10.1016/j.marmicro.2018.07.005.

Hashemi, Z., Biltekin, D., Greaves, A.M., Akçer-Ön, S., Ön, Z.B., Çağatay, M.N., Yetemen, Ö., 2026b. Holocene vegetation change, climate variability, and human impacts on the Seki Basin (southwestern Türkiye): evidence from a multiproxy approach. Palaeogeogr. Palaeoclimatol. Palaeoecol. 700 (15 October 2026), 114053. https://doi.org/10.1016/j.palaeo.2026.114053.

Hashemi, Z., Biltekin, D., Greaves, A.M., Akçer-Ön, S., Ön, Z.B., Zagvozdina, K., Yetemen, Ö., 2026a. Fire–climate–human interactions during the Holocene: insights from sediment records in southwestern Türkiye. Mediterranean Geosci. Rev. 8, 665–682. https://doi.org/10.1007/s42990-026-00247-0.

Hillbrand, M., van Geel, B., Hasenfratz, A., Hadron, P., Haas, J.N., 2014. Non-pollen palynomorphs show human- and livestock-induced eutrophication of Lake Nussbaumersee (Thurgau, Switzerland) since Neolithic times (3840 BC). Holocene 24 (5), 559–568. https://doi.org/10.1177/0959683614522307.

Hofmann, T.A., 2009. Plant Parasitic Asterinaceae and Microthyriaceae from the Neotropics (Panama). Faculty of Biological Sciences, Johann Wolfgang Goethe-University, Frankfurt am Main, Germany, p. 416. Ph.D. Thesis.

Hufnagel, L., Kocsis, M., 2011. Impacts of climate change on Lepidoptera species and communities. Appl. Ecol. Environ. Res. 9 (1), 43–72. https://doi.org/10.15666/aeer/0901_043072.

Iwamoto, A., Izumidate, R., Ronse De Craene, L.P., 2015. Floral anatomy and vegetative development in *Ceratophyllum demersum*: a morphological picture of an "unsolved" plant. Am. J. Bot. 102 (10), 1578–1589. https://doi.org/10.3732/ajb.1500124.

Jacobson, M.J., Flohr, P., Gascoigne, A., Leng, M.J., Sadekov, A., Cheng, H., Edwards, R. L., Tüysüz, O., Fleitmann, D., 2021. Heterogeneous late Holocene climate in the Eastern Mediterranean— the Kocain Cave Record from SW Turkey. Geophys. Res. Lett. 48, e2021GL094733. https://doi.org/10.1029/2021GL094733.

Keserci, F., Bayrakdar, C., Çılğın, Z., Evans, I.S., 2023. Modelling the form, distribution and paleoclimatic implications of former glaciers in the Teke Peninsula (Eastern Mediterranean, Turkey). Geomorphology 431, 108683. https://doi.org/10.1016/j.geomorph.2023.108683.

Khenzykhenova, F., Yoshida, K., Sato, T., Shchetnikov, A., Osipova, E., Danukalova, G., Ivanova, V., Simakova, A., Filinov, I., Semenei, E., Namzalova, O., Tumurov, E., Malikov, D., 2019. The Late Pleistocene Bokhan site (Fore-Baikal area, Russia) and its palaeoenvironmental reconstruction. Quat. Int. 534, 197–210. https://doi.org/10.1016/j.quaint.2019.04.023.

Kolbüken, M., Biltekin, D., Veldkamp, T., Arıkan, B., Acar, D., 2025. A 1100-years paleovegetation and paleoclimate record from western Türkiye linked to the North Atlantic Oscillation variability. Rev. Palaeobot. 343, 105417. https://doi.org/10.1016/j.revpalbo.2025.105417.

Kutluk, H., 2019. Palynomorphs from late Holocene sediments of the eutrophic Lake Manyas, NW Anatolia. Rev. Palaeobot. 269, 1–32. https://doi.org/10.1016/j.revpalbo.2019.06.001.

Leal, A., Martínez-Blanco, X., Beri, Á., Del Puerto, L., 2021. A combined catalog of non-pollen palynomorphs (NPPs) of fungal origin from soil and airborne samples of Uruguay. Rev. Palaeobot. 293, 104488. https://doi.org/10.1016/j.revpalbo.2021.104488.

Lee, C.M., Van Geel, B., Gosling, W.D., 2022. On the use of spores of coprophilous fungi preserved in sediments to indicate past herbivore presence. Quaternary 5 (3), 30. https://doi.org/10.3390/quat5030030.

Limaye, R.B., Kumaran, K.P.N., Nair, K.M., Padmalal, D., 2007. Non-pollen palynomorphs as potential palaeoenvironmental indicators in the Late Quaternary sediments of the west coast of India. Curr. Sci. 92 (10), 1370–1382. https://www.jstor.org/stable/24097765.

Lionello, P., Giorgi, F., Rohling, E., Seager, R., 2022. Mediterranean climate: past, present and future. In: Schroeder, K., Chiggiato, J. (Eds.), Oceanography of the Mediterranean Sea. Elsevier, pp. 41–91. https://doi.org/10.1016/B978-0-12-823692-5.00011-X.

Loughlin, N.J.D., Gosling, W.D., Montoya, E., 2018. Identifying environmental drivers of fungal non-pollen palynomorphs in the montane forest of the eastern Andean flank, Ecuador. Quat. Res. 89 (1), 119–133. https://doi.org/10.1017/qua.2017.73.

McCarthy, F.M.G., Pilkington, P.M., Volik, O., Heyde, A., Cocker, S.L., 2021. Non-pollen palynomorphs in freshwater sediments and their palaeolimnological potential and selected applications. Geological Society 511 (1), 121–150. https://doi.org/10.1144/SP511-2020-109. London, Special Publications.

Médail, F., Quezél, P., 1999. Biodiversity hotspots in the Mediterranean Basin: setting global conservation priorities. Conserv. Biol. 13 (6), 1510–1513. https://www.jstor.org/stable/2641976.

Medeanic, S., 2006. Freshwater algal palynomorph records from Holocene deposits in the coastal plain of Rio Grande do Sul, Brazil. Rev. Palaeobot. Palynol. 141 (1–2), 83–101. https://doi.org/10.1016/j.revpalbo.2006.03.012.

Medeanic, S., Silva, M.B., 2010. Indicative value of non-pollen palynomorphs (NPPs) and palynofacies for palaeoreconstructions: Holocene Peat. Brazil. Int. J. Coal Geol. 84, 248–257. https://doi.org/10.1016/j.coal.2010.08.015.

Menozzi, B.I., Zotti, M., Montanari, C., 2010. A non-pollen palynomorphs contribution to the local environmental history in the Ligurian Apennines: a preliminary study. Veg. Hist. Archaeobotany 19, 503–512. https://doi.org/10.1007/s00334-010-0271-1.

Mighall, T.M., Martínez Cortizas, A., Biester, H., Turner, S.E., 2006. Proxy climate and vegetation changes during the last five millennia in NW Iberia: pollen and non-pollen palynomorph data from two ombrotrophic peat bogs in the North Western Iberian Peninsula. Rev. Palaeobot. 141, 203–223. https://doi.org/10.1016/j.revpalbo.2006.03.013.

Miola, A., 2012. Tools for Non-Pollen Palynomorphs (NPPs) analysis: a list of Quaternary NPP types and reference literature in English language (1972–2011). Rev. Palaeobot. 186, 142–161. https://doi.org/10.1016/j.revpalbo.2012.06.010.

Momigliano, N., Greaves, A., Hodos, T., Aksoy, B., Brown, A., Kibaroğlu, M., Carter, T., 2011. Settlement history and material culture in southwest Turkey: report on the 2008–2010 survey at Çaltılar Höyük (northern Lycia). Anatol. Stud. 61, 61–121. https://doi.org/10.1017/S0066154600008796.

Montoya, E., Rull, V., Vegas-Vilarrébia, T., 2012. Non-pollen palynomorph studies in the Neotropics: the case of Venezuela. Rev. Palaeobot. Palynol. 186, 102–130.

Montoya, E., Rull, V., Vegas-Vilarrúbia, T., Corella, J.P., Giralt, S., Valero-Garcés, B., 2018. Grazing activities in the southern central Pyrenees during the last millennium as deduced from the non-pollen palynomorphs (NPP) record of Lake Montcortès. Rev. Palaeobot. Palynol. 254, 8–19.

Mudie, P.J., Leroy, S.A.G., Marret, F., Gerasimenko, N.P., Kholeif, S.E.A., Sapelko, T., Filipova-Marinova, M., 2011. Nonpollen palynomorphs: indicators of salinity and environmental change in the Caspian–Black Sea–Mediterranean corridor. In: Buynevich, I.V., Yanko-Hombach, V., Gilbert, A.S., Martin, R.E. (Eds.), Geology and Geoarchaeology of the Black Sea Region: Beyond the Flood Hypothesis. Geological Society of America, pp. 89–115. https://doi.org/10.1130/2011.2473(07.

Mudie, P.J., Marret, F., Rochon, A., Aksu, A.E., 2010. Non-pollen palynomorphs in the Black Sea corridor. Veg. Hist. Archaeobotany 19, 531–544. https://doi.org/10.1007/s00334-010-0268-9.

Non-pollen palynomorph database: https://non-pollen-palynomorphs.uni-goettingen.de/detail_type.html?Id=HdV-989).

Öntürk, T., Kutluk, H., 2022. Organic-walled sub-fossil invertebrate remains (non-pollen palynomorphs) from the Mid-Late Holocene (~5 kyrs) sediments of Lake Hazar (Eastern Anatolia, Turkey); implications for lake level changes. Palaeontograph. Abteilung 322, 153–181. https://doi.org/10.1127/pala/2022/0127.

Özhatay, N., Koçyiğit, M., Yüzbaşıoğlu, S., Gürdal, B., 2013. Mediterranean flora and its conservation in Turkey: with special reference to Monocot geophytes. Flora Mediterr. 23, 195–208. https://doi.org/10.7320/FlMedit23.195.

Quamar, M.F., 2015. Non-pollen palynomorphs from the late Quaternary sediments of southwestern Madhya Pradesh (India) and their palaeoenvironmental implications. Hist. Biol. 27 (8), 1070–1078. https://doi.org/10.1080/08912963.2014.933212.

Raczka, M.F., Bush, M.B., Folcik, A.M., McMichael, C.H., 2016. *Sporormiella* as a tool for detecting the presence of large herbivores in the Neotropics. Biota Neotropica 16 (1), e20150090. https://doi.org/10.1590/1676-0611-BN-2015-0090.

Raper, D., Bush, M., 2009. A test of *Sporormiella* representation as a predictor of megaherbivore presence and abundance. Quat. Res. 71 (3), 490–496. https://doi.org/10.1016/j.yqres.2009.01.010.

Reille, M., 1992. Pollen Et Spores D'Europe Et D'Afrique Du Nord. Laboratoire de botanique historique et de palynologie, Marseille, France, p. 543.

Revelles, J., Burjachs, F., Van Geel, B., 2016. Pollen and non-pollen palynomorphs from the Early Neolithic settlement of La Draga (Girona, Spain). Rev. Palaeobot. Palynol. 225, 1–20. https://doi.org/10.1016/j.revpalbo.2015.11.001.

Ricci, C., 2001. Dormancy patterns in rotifers. In: Sanoamuang, L., Segers, H., Shiel, R.J., Gulati, R.D. (Eds.), Rotifera IX. Springer, Netherlands, pp. 1–11. https://doi.org/10.1007/978-94-010-0756-6_1.

Rowe, P.J., Wickens, L.B., Sahy, D., Marca, A.D., Peckover, E., Noble, S., Özkul, M., Baykara, M.O., Millar, I.L., Andrews, J.E., 2020. Multi-proxy speleothem record of climate instability during the early last interglacial in southern Turkey. Palaeogeogr. Palaeoclimatol. Palaeoecol. 538, 109422. https://doi.org/10.1016/j.palaeo.2019.109422.

Rull, V., López-Sáez, J.A., Vegas-Vilarrúbia, T., 2008. Contribution of non-pollen palynomorphs to the paleolimnological study of a high-altitude Andean lake (Laguna Verde Alta, Venezuela). J. Paleolimnol. 40 (1), 399–411. https://doi.org/10.1007/s10933-007-9169-z.

Scott, L., 1992. Environmental implications and origin of microscopic *Pseudoschizaea Thiergart* and Frantz Ex R. Potonie emend, in Sediments. J. Biogeogr. 19 (4), 349–354. https://doi.org/10.2307/2845562.

Seeliger, M., Pint, A., Frenzel, P., Feuser, S., Pirson, F., Riedesel, S., Brückner, H., 2017. Foraminifera as markers of Holocene sea-level fluctuations and water depths of ancient harbours—A case study from the Bay of Elaia (W Turkey). Palaeogeogr. Palaeoclimatol. Palaeoecol. 482, 17–29. https://doi.org/10.1016/j.palaeo.2017.05.017.

Shumilovskikh, L., O'Keefe, J.M.K., Marret, F., 2021. An overview of the taxonomic groups of nonpollen palynomorphs. In: Marret, F., O'Keefe, J., Osterloff, P., Pound, M., Shumilovskikh, L. (Eds.), Applications of Non-pollen Palynomorphs: from Palaeoenvironmental Reconstructions to Biostratigraphy, vol 511. Geological Society, Special Publications, London, pp. 13–61. https://doi.org/10.1144/SP511-2020-65.

Shumilovskikh, L.S., Schlütz, F., Achterberg, I., Bauerochse, A., Leuschner, H.H., 2015. Nonpollen palynomorphs from mid-holocene peat of the raised bog Borsteler Moor (Lower Saxony, Germany). Stud. Quat. 32 (1), 5–18. https://doi.org/10.1515/squa-2015-0001.

Shumilovskikh, L.S., Schlütz, F., Lorenz, M., Tomaselli, M.B., 2019. Nonpollen palynomorphs notes: 3. Phototrophic loricate euglenoids in palaeoecology and the effect of acetolysis on Trachelomonas loricae. Rev. Palaeobot. Palynol. 270, 1–7. https://doi.org/10.1016/j.revpalbo.2019.06.017.

Shumilovskikh, L.S., Seeliger, M., Feuser, S., Novenko, E., Schlütz, F., Pint, A., Brückner, H., 2016. The harbour of Elaia: a palynological archive for human environmental interactions during the last 7500 years. Quat. Sci. Rev. 149, 167–187. https://doi.org/10.1016/j.quascirev.2016.07.014.

Shumilovskikh, L.S., Shumilovskikh, E.S., Schlütz, F., van Geel, B., 2022. NPP-ID: Non-Pollen Palynomorph Image Database as a research and educational platform. Veg. Hist. Archaeobotany 31 (3), 323–328. https://doi.org/10.1007/s00334-021-00849-8.

Shumilovskikh, L.S., van Geel, B., 2020. Nonpollen palynomorphs. In: Henry, A.G. (Ed.), Handbook for the Analysis of Microparticles in Archaeological Samples. Interdisciplinary Contributions to Archaeology. Springer, Cham, pp. 65–94. https://doi.org/10.1007/978-3-030-42622-4_4.

Simakova, A.N., Tesakov, A.S., Çelik, H., Frolov, P.D., Shalaeva, E.A., Sokolov, S.A., Trikhunkov, Y.I., Trifonov, V.G., Bachmanov, D.M., Latyshev, A.V., Ranjan, P.B., Gaydalenok, O.V., Syromyatnikova, E.V., Kovaleva, G.V., Vasilieva, M.A., 2021. Caspian-type dinocysts in NE Turkey mark deep inland invasion of the Akchagylian brackish-water basin during the terminal Late Pliocene. Quat. Int. 605–606, 329–348. https://doi.org/10.1016/j.quaint.2021.01.020.

Smol, J.P., Birks, H.J., Last, W.M., 2002. Tracking environmental change using Lake sediments: volume 3: terrestrial, algal, and Siliceous Indicators. Springer, Netherlands. https://doi.org/10.1007/0-306-47668-1.

Srivastava, J., Farooqui, A., Hussain, S.M., 2013. Climate-induced late-holocene ecological changes in Pichavaram estuary, India. Mar. Ecologist 34 (4), 474–483. https://doi.org/10.1111/maec.12048.

Tüfekcioğlu, İ., Tavşanoğlu, Ç., 2024. Growth form, regeneration mode, and vegetation type explain leaf trait variability at the species and community levels in Mediterranean woody vegetation. Nat. Ecol. Evol. 14 (3), e11145. https://doi.org/10.1002/ece3.11145.

Tunno, I., Mensing, S.A., 2017. The value of non-pollen palynomorphs in interpreting paleoecological change in the Great Basin (Nevada, USA). Quat. Res. 87 (3), 529–543. https://doi.org/10.1017/qua.2017.8.

Van Asperen, E.N., Perrotti, A., Baker, A., 2021. Coprophilous fungal spores: non-pollen palynomorphs for the study of past megaherbivores. Geol. Soc., London, Spec. Publ. 511 (1), 245–267. https://doi.org/10.1144/SP511-2020-41.

Van Geel, B., 2001. Nonpollen palynomorphs. Track. Environ. Change Using Lake Sed.: Terrestr. Algal, Silic. Indicat. 99–119.

Van Geel, B., Aptroot, A., 2006. Fossil ascomycetes in Quaternary deposits. Nova Hedwigia 82 (3), 313–330. https://doi.org/10.1127/0029-5035/2006/0082-0313.

Van Geel, B., Coope, G.R., Van Der Hammen, T., 1989. Palaeoecology and stratigraphy of the lateglacial type section at Usselo (The Netherlands). Rev. Palaeobot. Palynol. 60, 25–129. https://doi.org/10.1016/0034-6667(89)90072-9.

Van Geel, B., Van der Hammen, T., 1978. Zygnemataceae in quaternary Columbian sediments. Rev. Palaeobot. Palynol. 25 (5), 377–391. https://doi.org/10.1016/0034-6667(78)90021-0.

Vermoere, M., Bottema, S., Vanhecke, L., Waelkens, M., Paulissen, E., Smets, E., 2002. Palynological evidence for late-Holocene human occupation recorded in two wetlands in SW Turkey. Holocene 12 (5), 569–584. https://doi.org/10.1191/0959683602hl568rp.

Vermoere, M., Smets, E., Waelkens, M., Vanhaverbeke, H., Librecht, I., Paulissen, E., Vanhecke, L., 2000. Late Holocene environmental change and the record of human impact at Gravgaz near Sagalassos, Southwest Turkey. J. Archaeol. Sci. 27 (7), 571–595. https://doi.org/10.1006/jasc.1999.0478.

Voglmayr, H., Montes-Borrego, M., Landa, B.B., 2014. Disentangling *Peronospora* on *papaver*: phylogenetics, taxonomy, nomenclature and host range of downy mildew of Opium poppy (Papaver somniferum) and related species. PLoS One 9 (5), e96838. https://doi.org/10.1371/journal.pone.0096838.

Wick, L., Lemcke, G., Sturm, M., 2003. Evidence of Lateglacial and Holocene climatic change and human impact in eastern Anatolia: high-resolution pollen, charcoal, isotopic and geochemical records from the laminated sediments of Lake Van, Turkey. Holocene 13, 665–675. https://doi.org/10.1191/0959683603hl653rp.

Wolf, F.A., 1966. Fungus spores in East African lake sediments. Bull. Torrey Bot. Club 93, 104–113. https://doi.org/10.2307/2483751.
Wolf, F.A., 1967. Fungus spores in East African lake sediments. IV. Bull. Torrey Bot. Club 94, 480–486. https://doi.org/10.2307/2483565.
Yeloff, D., Mauquoy, D., Barber, K., Way, S., van Geel, B., Turney, C.M.S., 2007. Volcanic ash deposition and long-term vegetation change on subantarctic marion Island. Arctic Antarct. Alpine Res. 39, 500–511. https://doi.org/10.1657/1523-0430(06-040 [YELOFF]2.0.CO;2.